COMPANY BLUEPRINT · OPERATING MANUAL · EDITION 1

ZEBRAIN

The Complete Company Blueprint & Operating Manual

Strategy, finance, operations, architecture, design, growth, sales, risk and execution for Zebrain: the service that lets anyone make, test, fix and publish their own small language model on their own free GPU.

VERIFIED ASSUMPTION RECOMMENDATION ILLUSTRATIVE UNKNOWN · CONFIRM

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0.1

How to read this manual

This is a working manual for running Zebrain, not a pitch deck. Every chapter says what to do, why it matters, how to do it and who owns it. Because Zebrain is a pre-launch product, the most important skill in reading it is telling what is *true today* from what is *proposed*.

EVIDENCE LABELS USED THROUGHOUT

Label	Meaning	Source of truth
VERIFIED	Exists today. Checked directly against Zebrain's source code, its Supabase database, its pricing file, or a test run.	Codebase and database as of 26 Oct 2026 [1][2]
ASSUMPTION	A working premise needed to plan (jurisdiction, costs, exchange rate). Replace with real numbers as soon as they exist.	Stated next to each assumption
RECOMMENDATION	What this manual proposes the company should do. Not yet in place.	Reasoning given in place
ILLUSTRATIVE	Hypothetical numbers or examples that show a method. **Never actual company data.**	Worked examples only
UNKNOWN	Information this manual could not verify and must be confirmed before acting.	Listed in the confirmation register (§0.2)

External facts carry a numbered citation such as [4], resolved in the Bibliography. Where the only available source is a third-party aggregator rather than the vendor's own page, the text says so: re-check those before relying on them, especially prices.

0.2

What must be confirmed before implementation

The following could not be verified from the code, the database or public sources. Each one changes decisions later in this manual.

#	Item	Why it matters	Owner
1	UNKNOWN Legal entity, country of incorporation, founders and cap table	Tax, GST, contracts, payment gateway onboarding, and who can sign. This manual assumes an Indian private limited company (§9.1).	Founder
2	UNKNOWN Whether Kaggle's terms permit a commercial product that runs jobs on users' free Kaggle accounts	The free tier, which is Zebrain's main acquisition hook, depends on it. Highest-priority legal question (§9.4).	Founder + counsel
3	UNKNOWN Team size, roles and salaries	Budget, runway, org chart. Chapter 2 proposes a structure from a founder-only start.	Founder
4	UNKNOWN Cash in the bank and any funding	Runway. Chapter 3 uses an illustrative ₹10 lakh.	Founder / CFO
5	UNKNOWN GPU supplier and real cost per hour for paid L40S/H100 training and for hosting	Gross margin on every paid product (§3.6).	CTO
6	UNKNOWN Trademark availability of "Zebrain" in target classes and markets	Brand investment. Similar names exist in AI (§9.2).	Counsel
7	UNKNOWN Production domain, hosting account and deployment	The site runs locally today. README targets Vercel [1].	CTO
8	UNKNOWN Official logo file	No logo asset exists in the code. The cover uses the live site's pixel wordmark treatment (Silkscreen type), which is what the product shows today.	Design

0.3

Executive summary

Zebrain lets a person describe a job in plain words and get back a small language model of their own: trained on their own free Kaggle GPU, tested automatically, published to their own Hugging Face account, chat-ready with an API, and fixable when it answers wrong, without writing code. VERIFIED

₹0	4	7	0
Cost to the user to train a 1–3B model on Kaggle's free T4 VERIFIED	Run modes built: train, fix, serve, improve (Testify) VERIFIED	API routes, all server-side with row-level security VERIFIED	Paying customers today: product is pre-launch VERIFIED

Where the company stands

- **Product:** a working end-to-end pipeline (plan → data checks → baseline test → QLoRA training → retest → publish → chat/API → fix → Gemini-led "Testify" improvement) built with Next.js, Supabase, the Kaggle API, Hugging Face and Gemini [1]. VERIFIED
- **Evidence it works:** internal test models (an SQL assistant and a coding assistant, both 1.5B) were trained, published, served and called over the API. The coding model's exam score moved 73% → 78% across rounds while its held-out test score moved 38% → 42%: real but modest gains that show small models need better data, not just more rounds. VERIFIED
- **Not built yet:** payments, paid GPU training, always-on hosting, the Business service workflow, production deployment, a booking page (the link is a placeholder). VERIFIED

The bet

Fine-tuning is now cheap, but it is still a developer's activity: datasets, chat templates, LoRA settings, evaluation and hosting each take expertise. Zebrain's wager is that **owning a small, specialised model** (named, on-brand, on-topic, portable) is valuable to small businesses and makers who will never touch a notebook, and that a free first model on the user's own GPU is the cheapest possible way to reach them.

ILLUSTRATIVE

In the base-case model in Chapter 3, Zebrain reaches about ₹1.9 lakh monthly revenue by month 12 at a 65% gross margin, while still burning about ₹1.6 lakh per month. Break-even needs about ₹4.4 lakh monthly revenue at that cost base, so the plan needs about ₹25 lakh of outside funding or faster revenue by month 6. **These are hypothetical figures** that show the method. They are not forecasts of actual results.

The five decisions that matter most in the next 90 days

#	Decision	Recommendation
1	Is the Kaggle-hosted free tier legally safe as a commercial funnel?	RECOMMENDATION Get a written legal opinion before public launch. Build the paid GPU path in parallel so the business survives a "no" (§9.4).
2	Who is the first paying customer?	RECOMMENDATION Small Indian businesses needing an on-brand support or FAQ bot, sold through the done-for-you Business plan first. It needs no self-serve billing and teaches what customers actually ask (§1.3, §8).
3	How to make the models good enough?	RECOMMENDATION Invest in data generation (Gemini/Qwen-written datasets of ~2,000 examples, document upload), not more training rounds (§6.1).
4	Security of stored keys	RECOMMENDATION Encrypt Kaggle, Hugging Face and Gemini keys at rest and stop embedding long-lived tokens in Kaggle scripts before launch (§4.10).
5	Payments and hosting	RECOMMENDATION Launch with Razorpay (INR, GST invoices) and hosting on a single serverless GPU provider for ≤3B models (§4.8).

01

Executive Strategy & Business Model

What Zebrain is, who it serves, how it makes money, who it competes with, and what it must achieve at 3, 6, 12 and 36 months.

1.1 Company overview

1.2 Problem and value proposition

1.3 Customers and personas

1.4 Products and offerings

1.5 Business model and pricing

1.6 Market and competition

1.7 Goals and milestones

1.8 KPIs, risks and contingencies

1.1

Company overview

What it is. Zebrain is a web service that turns a plain-language description ("an assistant that writes PostgreSQL for our sales database") into a fine-tuned small language model owned by the user. VERIFIED The training runs on the user's own Kaggle account as a private GPU script, and the model is published to the user's own Hugging Face account as versions `v1`, `v2` ... with a model card that shows before-and-after scores [1].

RECOMMENDATION

MISSION

Give every business and maker a language model of their own that does one job well, answers in their name, and stays theirs.

VISION (2030)

Owning a specialised model is as ordinary as owning a website, and Zebrain is where people make, test, fix and run them.

RECOMMENDATION

VALUES

- **Yours, not ours.** Models, data and keys belong to the user. Portability by default.
- **Tested, not hoped.** Nothing ships without before/after scores.
- **Plain words.** No jargon a shop owner wouldn't use.
- **Honest about limits.** Say when a small model can't do something.
- **Frugal by design.** Free GPUs first, paid compute when it earns its cost.

Positioning statement

RECOMMENDATION

For **small businesses and independent builders** who want an AI assistant that knows their job and speaks as them, **Zebrain** is the **no-code model maker** that trains, tests and publishes a model you own, starting free on your own GPU. Unlike general chat apps such as ChatGPT or Gemini, a Zebrain model is trained on your examples, stays on topic, carries your name, and lives in your account, so you can take it anywhere.

1.2

Problem and value proposition

Problem today	What Zebrain does	Status
Fine-tuning needs a GPU, Python, and knowledge of LoRA, chat templates and evaluation.	A wizard plans the run. A pre-tested Python job does everything on the user's Kaggle T4.	VERIFIED
Picking good training data is hard.	"Let Gemini choose" searches Hugging Face, reads real sample rows, picks 1–3 open-licensed datasets and writes ~25 test questions.	VERIFIED
Fine-tuned models forget who they are ("I am Qwen, created by Alibaba Cloud").	Identity is written into the chat template and trained with identity, capability, greeting, off-topic and generic-system-prompt rows.	VERIFIED
Nobody knows if training helped.	Automatic tests before and after, graded by Gemini when a key is present. The best round is published.	VERIFIED
Fixing a wrong answer means starting over.	Fix runs turn corrections into ~20 examples each and continue from the latest adapter. Testify finds weak topics and trains on fresh practice.	VERIFIED
Using the model in an app needs hosting.	One-click chat plus `/ask` and `/predict` APIs on the user's Kaggle GPU for up to 11 hours, and a free static public page.	VERIFIED
Always-on hosting with an API key.	Planned Starter and Pro hosting plans.	PLANNED

1.3

Target customers and personas

None of these personas has been validated by customer interviews yet. ASSUMPTION §8.2 sets out the discovery interviews that must confirm or kill each one.

Persona	Who	Job to be done	Pain	Likely plan
Priya, shop owner	Runs a D2C brand with 2–10 staff. Answers the same WhatsApp and email questions daily.	"Answer return, size and delivery questions in our tone."	Generic bots give generic answers. Can't code.	Business, later Starter
Arjun, indie developer	Student or freelancer building apps and side projects.	"A small SQL/code model I own for my app, cheap to run."	API costs and lock-in. Fine-tuning is fiddly.	Free → Starter/Pro
Meera, ops lead	Ops at a 50–500 person company with internal docs and policies.	"An internal helpdesk model trained on our policies."	Data privacy worries with public chat apps.	Business
Dr. Rao, educator	Teacher or trainer making subject tutors.	"A tutor that stays on syllabus and refuses off-topic."	General bots wander off-topic.	Free → Starter

Priority use cases

1. Customer-support / FAQ bots with the company's name and policies (highest willingness to pay).
2. Text-to-SQL and code helpers for a specific stack (strongest technical results so far).
3. Internal knowledge helpers (needs document upload, not yet built).
4. Tutors and on-syllabus assistants.

1.4

Products, services and future offerings

Offering	Description	Status
Wizard + training	Describe job → base model (e.g. Qwen2.5-Coder-1.5B) → data → tests → train on Kaggle	LIVE (LOCAL)
Runs dashboard	Live stages, loss, tester scores, logs, resume from checkpoint	LIVE (LOCAL)
Chat + API	Gradio chat, `/ask` (key) and `/predict`, streaming, answer self-check and rewrite	LIVE (LOCAL)
Fix answers	Corrections from chat → targeted retraining, publish only if no regression	LIVE (LOCAL)
Testify with Gemini	Gemini exams the model, finds weak topics, writes practice, retrains, keeps the best	LIVE (LOCAL)
Paid training	L40S (≤14B) and H100 (15–30B) per GPU-hour	PRICED, NOT BUILT
Hosting	Free browser chat (≤3B), Starter, Pro, top-ups	PRICED, NOT BUILT
Business plan	Done-for-you model and website chatbot, priced in a meeting	DESIGNED, NOT BUILT
Dataset generation	LLM-written datasets with web search; document upload → topics → examples	DISCUSSED, PAUSED
Token gateway	Metered LLM tokens for data generation (e.g. 2M free trial, Pro bundle)	DISCUSSED, PAUSED

"Live (local)" means it works end to end on the development server against real Kaggle and Hugging Face accounts, but is not yet deployed to a public domain.

1.5

Business model, revenue streams and pricing

Zebrain has three revenue streams, all already priced in `lib/pricing.ts` [1]. VERIFIED None is switched on yet.

TRAINING (ONE-TIME PER RUN)

Tier	GPU	Model sizes	Price	Identity
Your own GPU	Kaggle free T4	1B–3B	Free	Small "Made with Zebrain" credit
Standard	L40S	Up to 14B	$2.75 / GPU-hour	Customer's own name
Large	H100	15B–30B	$7.50 / GPU-hour	Customer's own name

HOSTING (MONTHLY, PRICED AT THE 1.5B BENCHMARK AND SCALED BY MODEL SIZE ÷ 1.5)

Plan	1.5B	3B	7B	Includes
Free	₹0	₹0	–	Browser chat only, no API key, up to ~3B
Starter	₹499	₹999	₹2,329	1,000 messages, 1 API key, 10 req/min, 500-token replies
Pro	₹1,499	₹2,999	₹6,999	5,000 messages, 3 API keys, 30 req/min, 1,000-token replies
Top-up +1,000	₹299	₹599	₹1,399	One-off pack on Starter or Pro

Scaled prices follow the code's rule: price × size ÷ 1.5, rounded to the nearest amount ending in 9. Prices exclude GST. Hosting reply limits (500 / 1,000 tokens) predate the chat's new 1,536-token cap with automatic continuation. RECOMMENDATION Align the two before launch.

Business plan: book a meeting → Zebrain builds, tests and hosts the model plus a website chat box → price agreed per client. VERIFIED (copy on the pricing page). The booking URL is still a placeholder.

Every plan at a glance: live, priced and planned

Plan	What the customer gets	Price	Status
Free training	Train 1–3B models on own Kaggle T4. Chat + API on own GPU up to 11 h. Fix and Testify. "Made with Zebrain" identity.	₹0	LIVE (LOCAL)
Standard training	L40S, models up to 14B, own name on the model, credits for failed runs	$2.75 / GPU-hour	PRICED, NOT BUILT
Large training	H100, models 15–30B (a typical 30B run is $18–36)	$7.50 / GPU-hour	PRICED, NOT BUILT
Free hosting	Browser chat, no API key, models up to ~3B	₹0	PRICED, NOT BUILT
Starter hosting	1,000 messages/month, 1 API key, 10 req/min	₹499/mo at 1.5B (scales with size)	PRICED, NOT BUILT
Pro hosting	5,000 messages/month, 3 API keys, 30 req/min, 3 models	₹1,499/mo at 1.5B (scales with size)	PRICED, NOT BUILT
Message top-ups	+1,000 or +5,000 messages on Starter/Pro	₹299 per 1,000 at 1.5B	PRICED, NOT BUILT
Business	Zebrain builds, tests and hosts the model plus a website chat box, in the client's own name	Agreed in a meeting	DESIGNED, NOT BUILT
Data-generation Pro (token plan)	A Qwen-powered dataset builder with web search and document upload, metered in Qwen tokens: 20M tokens per model	$7 per model	DISCUSSED, PAUSED
Data-generation free trial	2M Qwen tokens to try dataset generation once	₹0	DISCUSSED, PAUSED
Annual plans	Starter/Pro paid yearly, 2 months free	10 × monthly	RECOMMENDATION
Agency / reseller	Margin on Business builds for agencies that resell them	Rev-share, to set	RECOMMENDATION

The token plan was discussed and paused: its database tables exist but are unused (§4.6). Estimated cost to Zebrain from earlier analysis: about $0.60–0.80 of model tokens per Pro model and about $0.20 per free trial, at roughly $0.03 / $0.13 per million input/output tokens for a Qwen Flash model. ASSUMPTION Re-check provider prices before launch.

Pricing strategy

- **Free as acquisition:** the free tier costs Zebrain almost nothing in compute, because the user's Kaggle quota pays. Kaggle reports around 30 free GPU hours per week per account [3] (third-party sources; confirm on Kaggle).
- **Value metric = messages:** customers understand messages, not tokens. Keep tokens internal.
- **Anchor against consumer chat apps:** the pricing page compares with ChatGPT Go (₹399), ChatGPT Plus (₹1,999), Google AI Plus (₹399) and Google AI Pro (₹1,950), as checked in October 2026 [12]. Zebrain is not cheaper per message than API models. It wins on ownership, on-brand behaviour and staying on topic, and the page says so honestly.
- RECOMMENDATION Add an annual plan (2 months free) once monthly churn data exists. Do not discount before then.

1.6

Market, competitors and SWOT

Market size. No credible, Zebrain-specific market sizing is available, and this manual will not invent one. UNKNOWN RECOMMENDATION Size the market bottom-up once discovery is done: (number of Indian SMBs with an online storefront or support inbox) × (share willing to pay for a custom bot) × (annual revenue per customer). Build each input from a sourced number, and show the sources in the investor deck.

Competitive landscape

Alternative	What it is	Price signal	Zebrain's difference
OpenAI fine-tuning	Fine-tune OpenAI models via API	~$0.80 per 1M training tokens for GPT-4.1 mini (third-party listing) [6]	Model stays on OpenAI. Developer workflow. Zebrain gives portable open weights.
Together AI	Per-token LoRA/full fine-tuning plus dedicated endpoints	Billed per token by size bracket [7]	Developer-first. Zebrain is no-code with a built-in tester and fixer.
Hugging Face AutoTrain	No-code training on HF compute	Per-minute hardware billing, small jobs free [8]	Generic trainer. No identity, off-topic, tester or fix loop.
ChatGPT / Gemini apps	General assistants with custom instructions	₹399–₹1,999 / month [12]	Not owned, not trained on your answers, not on-brand.
DIY (Colab + Unsloth/PEFT)	Notebooks and tutorials	Free compute, high skill	Same free GPU, none of the skill.

Strengths
Working end-to-end pipeline with automatic testing and fixing
Near-zero compute cost for the free tier
Identity and on-topic training solves a real annoyance
User owns everything (lock-in fear removed)

Weaknesses
1–3B models have a low ceiling on hard tasks (test score 42%)
Dependent on Kaggle, Hugging Face and Gemini policies
Onboarding needs three accounts (Kaggle with phone verification, HF, Gemini)
No payments, hosting or production deployment yet

Opportunities
SMB support bots in Indian languages and tone
Privacy-sensitive buyers who want their own weights
Agencies reselling done-for-you bots
Better small open models arriving every few months

Threats
Kaggle terms or quota changes (existential for free tier)
Big platforms adding one-click custom models
Falling API prices make "good enough" generic bots cheaper
Data-protection obligations under India's DPDP Rules [4]

1.7

Business goals and milestones

Horizon	Goals	Exit criteria (measurable)
3 months	Legal clarity on Kaggle use. Production deploy. Keys encrypted. Payments (Razorpay). 10 discovery interviews. First 3 Business pilots.	Site live on own domain. 3 signed pilots. 100 sign-ups. 30 models trained by users.
6 months	Hosting (≤3B) launched. Paid L40S training. Dataset generation from documents.	₹50k MRR. 40% of trained models pass tests on first run. Under 5% failed runs caused by Zebrain.
12 months	Self-serve Starter/Pro at scale. Agency/reseller programme. 2 hires.	₹1.9 lakh MRR (illustrative base case). Logo churn under 6% a month. NPS 40+.
36 months	Category leader for owned small models in India and other price-sensitive markets. Enterprise private deployments.	Break-even or funded growth. 2,000+ hosted models. Gross margin 70% or more.

1.8

KPIs, risks and contingency plans

North-star metric: **models in active use**, meaning hosted or Kaggle-served models that answered at least 50 messages in the last 30 days. It captures value delivered (a model good enough to use), not just sign-ups. Full KPI definitions are in Chapter 10.

Top risk	Likelihood	Impact	Contingency
Kaggle disallows or blocks this use	Medium	Severe	Paid GPU training ready. Offer Colab/own-GPU export of the job script. Free trial credits on Zebrain GPUs.
Model quality too low for paying customers	High	High	Lead with Business plan (humans curate data). Default to 3B where quality matters. Invest in data generation.
Leak of stored user keys	Low–Med	Severe	Encryption at rest, scoped tokens, rotation prompts, incident plan (§4.10, §9.6).
Gemini free-tier limits or terms	Medium	Medium	Rule-based fallback already exists. Paid Gemini tier for Zebrain-run jobs. Alternative LLM provider.
Founder bandwidth	High	High	Strict 90-day scope (Chapter 11). First hire is a full-stack/ML engineer.

Decision framework for new features

```
Priority score = (Reach × Impact × Confidence) ÷ Effort  ·  Reach = users per quarter · Impact 0.25-3 · Confidence 50-100% · Effort = person-weeks
```

Any feature that touches user keys, user data or model publishing also needs a security review sign-off (§6.4), whatever its score.

02

Company Structure & Internal Operations

How a founder-led Zebrain should be organised now, who it hires next, and the operating procedures that keep a small team fast without dropping things.

2.1 Organisation and ownership

2.2 Hiring roadmap

2.3 Operating rhythm and decisions

2.4 Standard operating procedures

2.5 Vendors, partners and outsourcing

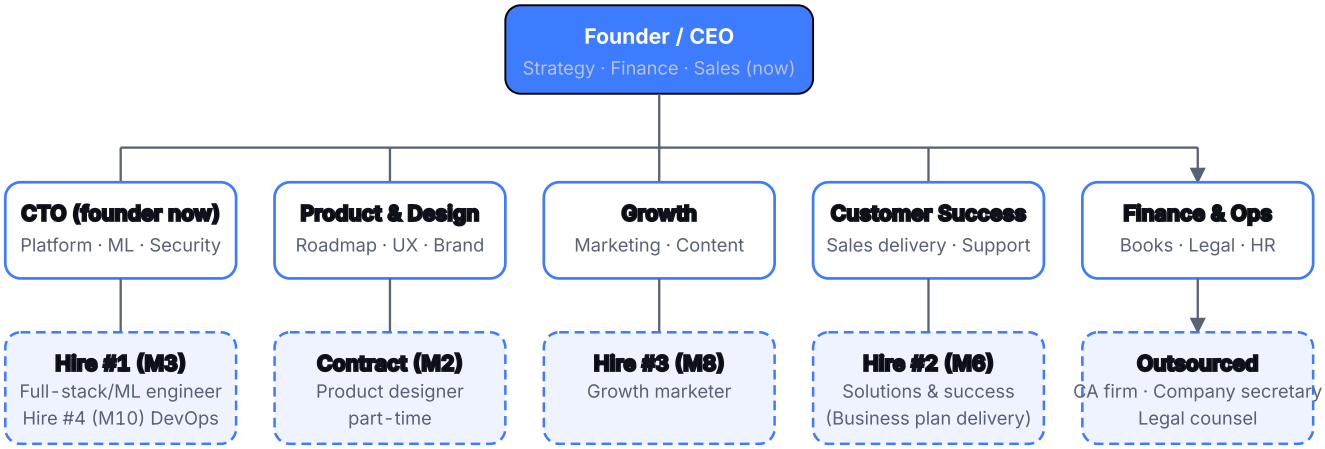
2.6 Remote work and onboarding

2.7 Legal documents and IP hygiene

2.1

Organisational structure and ownership

The current team size is not recorded anywhere this manual could check. UNKNOWN The structure below assumes **one founder today**, growing to about six people in 12 months. Each box is a *role*, not a person: at first the founder holds most of them, and the point of the chart is that every responsibility has exactly one named owner. RECOMMENDATION



Solid boxes: functions (one owner each). Dashed boxes: proposed hires and outsourced partners, with target month.

FIG 2.1 Proposed organisation chart, months 0–12. RECOMMENDATION

Responsibility matrix (RACI)

Responsibility	CEO	CTO	Product	Growth	Success	Finance
Strategy, fundraising, investor updates	A/R	C	C	C	I	C
Roadmap and prioritisation	A	C	R	C	C	I
Training job, tester, model quality	I	A/R	C	I	C	I
Security, keys, incidents	I	A/R	I	I	C	I
Brand, website, content	C	I	C	A/R	I	I
Business-plan delivery	A	C	I	I	R	I
Support queue and SLAs	I	C	I	I	A/R	I
Books, GST, payroll, compliance	A	I	I	I	I	R

R = does the work · A = accountable, makes the final call · C = consulted · I = informed. Exactly one A per row.

2.2

Hiring roadmap and key job descriptions

When	Role	Why then	Monthly cost (₹)
M2	Product designer (contract, 40%)	Onboarding is the main conversion risk (three external accounts).	40,000
M3	Full-stack / ML engineer	Hosting, payments and paid GPUs can't all be built by one person.	1,20,000
M6	Solutions & customer success	Business pilots need a dedicated owner.	70,000
M8	Growth marketer	After the message and ICP are validated, not before.	80,000
M10	DevOps / SRE (part-time to start)	Hosting reliability once paying customers depend on it.	60,000

ILLUSTRATIVE Costs are placeholder Bengaluru-market figures for planning only. Benchmark against current offers before hiring.

JD · Full-stack / ML engineer

Mission: make every Zebrain model train, test and serve reliably at lower cost.

- Own the Python job (`zebrain_train.py`): data checks, QLoRA, tester, serving.
- Build paid GPU training and hosting.
- Next.js/TypeScript API routes, Supabase, CI.

Must: PyTorch, Hugging Face transformers/PEFT, TypeScript, SQL. **Nice:** vLLM/TGI, GPU cost tuning.

90-day outcome: hosting for ≤3B models live with under 1% failed requests.

JD · Solutions & customer success

Mission: turn Business-plan clients into happy, renewing customers.

- Run discovery calls, collect FAQs and documents, write test questions with the client.
- Use Zebrain to build, fix and Testify client models, then go live.
- Own onboarding, support SLAs and renewals.

Must: clear writing, client handling, comfort with data. **Nice:** SQL, chatbot deployments.

90-day outcome: 5 live clients, average CSAT 4.5/5 or higher.

2.3

Operating rhythm, communication and decisions

Ritual	Cadence	Purpose and output
Daily async check-in	Daily, written	Yesterday / today / blocked, in one Slack thread. No meeting.
Weekly planning	Monday, 45 min	Review KPIs (Appendix A17), commit the week's top 3 outcomes per owner.
Demo & retro	Friday, 30 min	Show shipped work on a real model. One improvement for next week.
Monthly business review	First Tuesday	P&L, runway, funnel, model quality, risks. Written memo first.
Quarterly planning	Quarterly	OKRs, budget re-forecast, hiring decisions.

Decision rules

- **Two-way doors** (reversible: copy, UI, prompts): the owner decides and tells the team.
- **One-way doors** (pricing, data handling, contracts, hires, anything touching user keys): a written one-page decision record (context, options, decision, owner, review date), stored in `/decisions` in the repo.
- **Approvals:** spend under ₹25,000 needs the budget owner. ₹25,000–₹2,00,000 needs CEO approval. Over ₹2,00,000 needs CEO plus a second signatory or a board note (§3.10).

Documentation and tools

RECOMMENDATION One home for each kind of knowledge: code and technical decisions in the Git repository; specs, SOPs and meeting notes in a shared docs space; tasks in a single tracker (Linear or GitHub Projects); customer conversations in the CRM (§8.3). If it isn't written down in its home, it didn't happen.

Project management and accountability

Every task has one owner, a due date and a definition of done. Work is planned in 2-week cycles (§6.2). Slipped commitments are discussed at the retro, without blame, and the plan is re-cut rather than silently extended.

2.4

Standard operating procedures

SOP	Trigger → steps → done	Owner
New user activation check	Sign-up without a trained model after 48 h → check whether Connect failed (Kaggle phone verification is the usual blocker) → send a targeted help email → log the reason in the CRM.	Success
Failed run triage	Run ends in error → read the stage and message in `/runs` → classify (user account / data / Zebrain bug / Kaggle) → if Zebrain's fault, file a bug at the right severity (§6.5) and credit the user once paid runs exist.	CTO
Business client delivery	Signed SOW → kickoff → collect data → write 25+ acceptance questions with the client → train → Testify → client UAT → go live → 30-day review.	Success
Release	See release checklist (§6.3).	CTO
Key/security incident	See incident checklist (Appendix A15).	CTO
Month-end close	See §3.5.	Finance

2.5

Vendors, procurement, partnerships and outsourcing

Vendor	Used for	Status	Risk and mitigation
Kaggle (Google)	Free T4 GPU for user jobs, via users' own accounts	VERIFIED	Terms and quota changes. Get a legal opinion. Keep the paid GPU fallback.
Hugging Face	Model repos, private state repo, static chat Space	VERIFIED	Gradio/Docker Spaces need PRO. Static Space used instead.
Google Gemini API	Planning, data review, grading, Testify	VERIFIED	Free-tier data terms [9], overload/503s. Probing and fallback built.
Supabase	Auth, Postgres, row-level security (Mumbai region)	VERIFIED	Move to a paid plan with backups and PITR before launch.
Vercel	Web hosting (planned in README)	ASSUMPTION	Function timeouts. `/api/plan` needs up to 300 s.
Razorpay / Stripe	Payments	RECOMMENDATION	Razorpay for INR plus GST invoices. Stripe for international customers later.
GPU cloud (TBD)	Paid training and hosting	UNKNOWN	Shortlist 3 providers. Compare $/hour, availability, India latency, DPA.

Procurement rule: any vendor that will receive user data needs a signed data processing agreement (DPA) and an entry in the vendor register (name, data shared, region, DPA link, owner, renewal date). RECOMMENDATION

2.6

Remote operations and onboarding

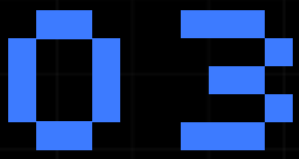
RECOMMENDATION Remote-first with a monthly in-person day. New-hire onboarding:

- [ ] Day 0: signed offer, NDA and IP assignment (§2.7), company accounts, password manager, 2FA everywhere.
- [ ] Day 1: read this manual's chapters 1, 4 and 6. Set up the dev environment (`npm install` , `npm run dev` , Python job tests).
- [ ] Day 2–3: train, chat with, fix and Testify your own model end to end as a user. Write down every point of friction.
- [ ] Week 1: ship one small fix to production with a reviewed PR.
- [ ] Day 30 / 60 / 90: check-ins against the role's 90-day outcome.

2.7

Legal documentation, IP and contracts

- **IP assignment:** every founder, employee and contractor signs an invention/IP assignment, so all of Zebrain's code (including `zebrain_train.py`) belongs to the company. RECOMMENDATION Founders assign pre-incorporation code to the company at incorporation.
- **Third-party licences:** keep a register of model and dataset licences. Base models carry their own licences (e.g. Qwen2.5-Coder-1.5B-Instruct is Apache-2.0 [11]; Llama fine-tunes must carry "Built with Llama" and a `llama-` name prefix, which the job already does [10]). The planner prefers open-licensed datasets. VERIFIED
- **Contract set:** Terms of Service, Privacy Policy, DPA, Business-plan MSA + SOW, contractor agreement, employment agreement, NDA. See Chapter 9.



Accounting, Finance & Financial Management

A practical finance operating system for a pre-revenue Indian SaaS company: books, revenue rules, closing, statements, unit economics, forecasting, controls and tax.

3.1 Assumptions and principles

3.2 Chart of accounts

3.3 Bookkeeping and transactions

3.4 Revenue recognition

3.5 Payables, receivables, closing

3.6 Unit economics

3.7 Financial statements

3.8 Cash, runway and scenarios

3.9 Fundraising and equity

3.10 Controls and fraud prevention

3.11 Tax and compliance

3.1

Assumptions and principles

UNKNOWN

No company financials exist in any source this manual could check: no bank balance, revenue, payroll or entity details. **Every number in this chapter is illustrative.** It exists to show the method and to be replaced by real figures.

Planning assumption	Value	Basis
Entity and jurisdiction	Indian Pvt Ltd	ASSUMPTION INR pricing, GST copy and a Mumbai database region in the product [1][2]
Exchange rate	$1 = ₹88	ASSUMPTION Planning rate only. Update monthly.
Paid GPU cost: L40S / H100	$1.10 / $2.60 per hour	ASSUMPTION All-in (compute, storage, idle). Confirm with the chosen provider.
Hosting compute per model per month: Starter / Pro (1.5B)	₹150 / ₹400	ASSUMPTION Shared serverless GPU with scale-to-zero
Payment gateway fee	2% of revenue	ASSUMPTION Typical domestic card/UPI pricing. Confirm with the gateway.
Starting cash	₹10,00,000	ILLUSTRATIVE
Monthly logo churn (hosting)	6%	ASSUMPTION Replace after 3 months of data
Blended CAC (self-serve hosting)	₹1,500	ASSUMPTION

Principles: books are kept on the accrual basis from day one. One bank account is used for business only. Every transaction has a document. The books close by the 7th working day of each month. A chartered accountant (CA) reviews them quarterly, and they are audited annually as Indian company law requires ASSUMPTION: CONFIRM WITH CA.

3.2

Chart of accounts

Code	Account	Use at Zebrain
1000 Assets		
1010	Bank: operating (INR)	All receipts and payments
1020	Payment gateway clearing	Razorpay balance before settlement
1100	Accounts receivable	Business-plan invoices
1200	GST input tax credit	GST paid on vendor bills
1210	TDS receivable	Tax deducted by clients
1300	Prepaid expenses	Annual SaaS tools, domain
1500	Computer equipment	Laptops (depreciated)
2000 Liabilities		
2010	Accounts payable	Vendor bills due
2100	GST output payable	GST charged to customers
2110	TDS payable	Deducted from contractor/salary payments
2200	Deferred revenue	Unused top-ups, prepaid plans and setup fees
2300	Customer credits	GPU-hour credits for failed paid runs
2400	Payroll liabilities	Salaries, PF and professional tax payable
3000 Equity		
3010	Share capital	Founder and investor shares
3020	Securities premium	Investor money above face value
3100	Retained earnings	Accumulated profit or loss
4000 Revenue		
4010	Hosting subscriptions	Starter, Pro
4020	Message top-ups	Recognised when used or on expiry
4030	Paid training (GPU-hours)	L40S and H100 runs
4040	Business plan: setup	Done-for-you builds
4050	Business plan: managed hosting	Monthly retainers
5000 Cost of revenue		
5010	GPU compute: training	Paid runs only (free runs cost Zebrain nothing)
5020	GPU compute: hosting	Inference
5030	LLM API usage	Gemini/Qwen for planning, grading and data
5040	Payment processing fees	Gateway charges
5050	Business delivery labour	Contractor time on client builds
6000 Operating expenses		

Code	Account	Use at Zebrain
6010	Salaries and benefits	Employees
6020	Contractors	Design, freelance engineering
6100	Software and infrastructure	Supabase, Vercel, GitHub, Google Workspace
6200	Marketing and advertising	Ads, content, events
6300	Legal and professional	CA, CS, counsel, trademark
6400	Travel and meetings	Client visits
6500	Bank charges and FX	Wire fees, currency loss
6900	Depreciation	Equipment

3.3

Bookkeeping procedures and transaction recording

1. **Capture.** Every bill and receipt is forwarded to a single finance inbox and attached in the accounting software (Zoho Books or Tally are common choices for Indian GST). RECOMMENDATION
2. **Code.** Assign the account (§3.2), the GST rate and the vendor's GSTIN, and tag the cost to product or overhead.
3. **Reconcile weekly.** Match the bank feed and the Razorpay settlement reports to entries. The gateway clearing account (1020) should net to zero after each settlement.
4. **Invoices.** Self-serve customers get automatic GST invoices from the gateway or billing system. Business clients get numbered invoices (`ZB/2026-27/0001`), due in 15 days.
5. **Refunds.** Refunds are issued against the original payment, with a credit note referencing the invoice. Failed paid runs caused by Zebrain get GPU-hour credits (account 2300) instead of cash, as the pricing page promises. VERIFIED (pricing copy)
6. **Reimbursements.** Claimed monthly with receipts, approved by the budget owner and paid with payroll.
7. **Payroll.** Processed monthly with TDS on salaries, PF and professional tax as applicable, using a payroll service.

ILLUSTRATIVE Journal entry for a ₹499 Starter subscription paid by UPI (GST 18% added):

```
Dr 1020 Gateway clearing        588.82
    Cr 4010 Hosting subscriptions        499.00
    Cr 2100 GST output payable            89.82
Dr 5040 Payment processing fees  11.78   (2% of 588.82)
    Cr 1020 Gateway clearing              11.78
On settlement: Dr 1010 Bank 577.04 / Cr 1020 Gateway clearing 577.04
```

3.4

Revenue recognition and expense categorisation

Stream	When revenue is recognised
Monthly hosting plans	Evenly over the service month. Annual plans are deferred and recognised monthly.
Message top-ups	Deferred when sold. Recognised as messages are used, with any remainder recognised on expiry (state an expiry, e.g. 12 months, in the terms).
Paid training	When the run completes and the model is delivered. Prepaid hours not yet used stay deferred. Credits issued for failures are a liability, not revenue.
Business setup fee	On delivery milestones agreed in the SOW (e.g. 50% at a model that passes acceptance tests, 50% at go-live).
Business managed hosting	Monthly, as delivered.

Indian accounting standards (AS or Ind AS, depending on company size) govern the details. Confirm the treatment, especially for top-up expiry, with the CA. CONFIRM

Expense rule of thumb: anything that scales with each unit served (GPU, LLM API, gateway fees, delivery labour) is *cost of revenue*. Everything else is *operating expense*. That split is what makes the gross margin meaningful.

3.5

Payables, receivables and the closing calendar

Accounts payable

- Pay vendors on net-15/30 terms, in two weekly payment runs.
- Check the GSTIN and the bill against the PO or contract before paying.
- Deduct TDS where applicable and deposit it by the due date.

Accounts receivable

- Business invoices are due in 15 days.
- Reminders at day +3, +10 and +20. Service pauses at +30 (stated in the MSA).
- Track DSO monthly. Target under 30 days.

CLOSING CALENDAR

Cycle	Steps	Deadline
Monthly	Reconcile bank and gateway → record accruals (GPU usage not yet billed) → move deferred revenue → payroll entries → review the P&L against budget → write the 1-page finance memo (Appendix A4)	Working day 7
Monthly tax	GST returns (GSTR-1, GSTR-3B) and TDS deposit, per current due dates	Statutory dates
Quarterly	TDS returns. CA review. Re-forecast (§3.8). Investor update (§3.9).	Quarter end + 30 days
Annual	Statutory audit, financial statements, income tax return, ROC annual filings, GST annual return where applicable	Statutory dates

3.6

Pricing, margins and unit economics

```
Gross margin = (Revenue − Cost of revenue) ÷ Revenue
Contribution margin = Price − variable cost per unit (compute + API + gateway)
CAC = (Sales + marketing spend in period) ÷ new paying customers in period
LTV = (ARPU × gross margin %) ÷ monthly churn  ·  Payback (months) = CAC ÷ (ARPU × GM %)
Break-even revenue = Fixed operating costs ÷ gross margin %
```

WORKED EXAMPLE · STARTER CUSTOMER, 1.5B MODEL ILLUSTRATIVE

Line	₹ per month	Note
Starter plan	499.00	From pricing file VERIFIED
Top-ups (25% of customers buy one)	74.75	0.25 × ₹299
ARPU	**573.75**	
Hosting compute	−150.00	Assumption
Top-up compute	−22.50	0.25 × ₹90
Gateway fees (2%)	−11.48	
Contribution per customer per month	**389.77**	**68% margin**
Expected lifetime at 6% monthly churn	16.7 months	1 ÷ 0.06
LTV	**6,496**	389.77 ÷ 0.06
CAC	1,500	Assumption
LTV : CAC · Payback	**4.3 : 1 · 3.8 months**	**Healthy if real churn is close to 6%**

CONTRIBUTION PER PAID TRAINING HOUR ILLUSTRATIVE

Tier	Price (₹/h)	GPU cost	Gateway	Contribution	Margin
Standard (L40S)	242.00	96.80	4.84	140.36	58%
Large (H100)	660.00	228.80	13.20	418.00	63%

Free Kaggle runs carry no compute cost for Zebrain. Their only variable cost is Gemini usage when Zebrain's server key is used as the fallback. RECOMMENDATION Require users to bring their own Gemini key for free runs, as the Connect page already allows, so this cost stays near zero.

3.7

Financial statements: month-12 base case

PROFIT & LOSS · MONTH 12 ILLUSTRATIVE

Line	₹
Starter (60 × 499)	29,940
Pro (20 × 1,499)	29,980
Top-ups (15 × 299)	4,485
Training L40S (120 h)	29,040
Training H100 (20 h)	13,200
Business (2 × 40,000)	80,000
Revenue	**1,86,645**
Hosting compute	−18,350
Training GPU	−16,192
Business delivery	−24,000
Gateway + LLM API	−6,733
Gross profit (65.0%)	**1,21,370**
People (founder, engineer, designer)	−2,10,000
Software & infra	−12,000
Marketing	−40,000
Legal, accounting, misc	−23,000
Operating loss	**−1,63,630**

BALANCE SHEET · END OF MONTH 12 ILLUSTRATIVE

Line	₹
Cash	14,70,000
Receivables	80,000
GST input credit	18,000
Total assets	**15,68,000**
Payables	25,000
GST/TDS payable	38,000
Deferred revenue	35,000
Share capital + premium	35,00,000
Retained earnings (year-1 loss)	−20,30,000
Liabilities + equity	**15,68,000**

Assumes ₹10 lakh founder capital plus a ₹25 lakh angel round in month 6 (§3.8), and founder-owned laptops (no fixed assets). The year-1 loss is the total of the quarterly plan in §3.8.

CASH FLOW STATEMENT · YEAR 1 ILLUSTRATIVE

Line	₹
Net loss	−20,30,000
Increase in receivables and GST input credit	−98,000
Increase in payables, GST/TDS payable and deferred revenue	98,000
Cash from operations	**−20,30,000**
Cash from investing	0
Equity raised (founder ₹10L + angels ₹25L)	35,00,000
Closing cash (= balance sheet cash)	**14,70,000**

The three statements tie: closing cash matches the balance sheet, and retained earnings equal the year's net loss. The month-end checklist (Appendix A4) tests the same ties on the real books.

3.8

Cash, burn, runway and scenario planning

```
Net burn = cash out − cash in (per month)  ·  Runway (months) = cash ÷ average net burn of last 3 months
Working capital = current assets − current liabilities
```

YEAR-1 PLAN BY QUARTER · BASE CASE ILLUSTRATIVE

Quarter	Revenue	Cost of revenue	Opex	Net	Cash at end
Q1 (M1–3): setup, pilots	30,000	-10,000	-4,00,000	-3,80,000	6,20,000
Q2 (M4–6): hosting, paid GPU; angels raised	1,50,000	-55,000	-6,00,000	-5,05,000	26,15,000
Q3 (M7–9)	3,60,000	-1,30,000	-8,55,000	-6,25,000	19,90,000
Q4 (M10–12)	5,20,000	-1,85,000	-8,55,000	-5,20,000	14,70,000
Year 1	**10,60,000**	**-3,80,000**	**-27,10,000**	**-20,30,000**	

Without the ₹25 lakh round, cash runs out in month 7. **Decision rule:** if the round hasn't closed by month 5, freeze hiring, cut marketing to ₹10,000 per month and focus on Business clients, which pay upfront setup fees. RECOMMENDATION

SCENARIOS AT MONTH 12 ILLUSTRATIVE

Driver	Downside	Base	Upside
Hosting customers (Starter + Pro)	30	80	180
Business clients	1	2	5
Paid training GPU-hours / month	40	140	400
Monthly revenue (approx.)	₹80k	₹1.87 lakh	₹4.6 lakh
Kaggle free tier	Blocked at M6	Available	Available
Response	Bootstrap mode, Business-only	Plan as written	Hire the growth marketer early (M6)

Budget management: a 12-month budget by account (Appendix A2) is approved each quarter. Variances over 10% and ₹20,000 are explained in the monthly memo.

3.9

Fundraising, investor reporting and equity

- **Instruments:** Indian pre-seed rounds commonly use equity or compulsorily convertible preference shares (CCPS). Valuation, FEMA rules for foreign investors and angel-tax provisions all need a CA or lawyer. CONFIRM WITH COUNSEL
- **Cap table hygiene:** keep a single source cap table (founders, ESOP pool, investors, fully diluted %). RECOMMENDATION Create a 10% ESOP pool before the first priced round, with standard vesting of 4 years and a 1-year cliff.
- **Investor update** (monthly, 1 page): headline, KPIs (MRR, models in active use, burn, runway), wins, misses, asks.
- **Data room:** incorporation documents, cap table, financials, this manual, product demo, security summary, key contracts.

- **Non-dilutive options:** Startup India (DPIIT) recognition and state startup schemes, and cloud credit programmes for GPUs [17].

3.10

Financial controls, approval limits and fraud prevention

Control	Rule
Approval limits	Under ₹25k: budget owner · ₹25k–₹2L: CEO · over ₹2L: CEO plus a second signatory · new recurring vendor: CEO
Segregation of duties	The person who adds a vendor never releases its payment. Bank payments need maker plus checker in net banking.
Vendor master changes	Bank-detail changes are confirmed by a phone call to a known number (the most common invoice fraud).
Cards	Company cards per person with monthly limits. Receipts within 7 days.
Cloud spend	Billing alerts at 50%, 80% and 100% of budget on every GPU and API account. Hard caps where available.
Refunds and credits	Over ₹5,000 needs CEO approval. All refunds and credits are logged with a reason code.
Audit trail	The accounting system's audit log stays on. Documents are retained for the statutory period (8 years under Indian company law is common; confirm).

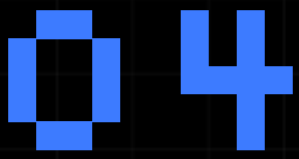
3.11

Tax and regulatory obligations (India)

CONFIRM WITH A CA

This section lists the obligations to discuss with a chartered accountant. It is not tax advice, and rates and thresholds change.

Area	What applies, as generally understood
GST	Software and SaaS services are generally taxed at 18% GST [5]. Registration depends on turnover thresholds and supply type. Many SaaS companies register early so they can collect GST and claim input credit.
Exports	Services to foreign customers can be zero-rated by filing a Letter of Undertaking (LUT, Form GST RFD-11) each year. Foreign payment must arrive within the prescribed period [5].
Imports of services	Paying foreign vendors (GPU clouds, Supabase, Vercel, Gemini) may trigger GST under reverse charge for a registered business. Track these bills separately.
TDS	Deduct TDS on contractor, professional and rent payments and on salaries, deposit it monthly and file quarterly returns.
Income tax	Domestic companies pay corporate tax under the regime they choose. Startup tax holidays need separate approval.
Company law	Statutory auditor appointment, board meetings, annual filings with the Registrar of Companies, director KYC [17].
Payroll	Professional tax (state), PF/ESI once headcount thresholds apply, labour-law registrations (Shops & Establishments).



Technical Architecture & Backend

The system as it exists in the codebase today, and the architecture proposed for launch: hosting, payments, security hardening and scale.

4.1 Verified vs proposed

4.2 System architecture

4.3 Frontend

4.4 Auth and permissions

4.5 Training job and core logic

4.6 Database

4.7 API design

4.8 Serving and hosting

4.9 Infrastructure and delivery

4.10 Security

4.11 Performance and cost

4.12 Privacy and data lifecycle

4.13 AI/LLM infrastructure

4.1

Verified architecture vs proposed architecture

Layer	Exists today VERIFIED	Proposed for launch RECOMMENDATION
Web app	Next.js 14.2 App Router, React 18.3, TypeScript 5.6, Tailwind 3.4. Runs locally (dev server).	Deployed to Vercel (or similar) with preview deployments per PR and a custom domain.
Auth and data	Supabase Auth (email/password; Google pending configuration) and Postgres with row-level security, Mumbai region.	Paid plan, daily backups plus point-in-time recovery, encrypted secrets columns.
Training compute	The user's own Kaggle account: a private GPU script on a T4, pushed via the Kaggle API.	Plus a Zebrain-run GPU queue (L40S/H100) for paid runs and larger models.
Model storage	User's Hugging Face: merged model repo (tagged per version) plus private state repo.	Unchanged. Optional Zebrain-managed storage for Business clients.
Serving	Gradio app on the user's Kaggle GPU (share link, up to 11 h), `/ask` and `/predict`, streaming with an answer checker, plus a static HF Space page.	Always-on hosting: serverless GPU inference (vLLM) behind a Zebrain API gateway with keys, quotas and metering.
LLM services	Gemini (newest stable flash model, probed for availability) for planning, data review, grading and Testify.	Provider abstraction (Gemini plus a second provider), token metering per user, prompt registry.
Payments	None.	Razorpay subscriptions, top-ups and GPU-hour credits, with webhooks into Postgres.

4.2

High-level system architecture

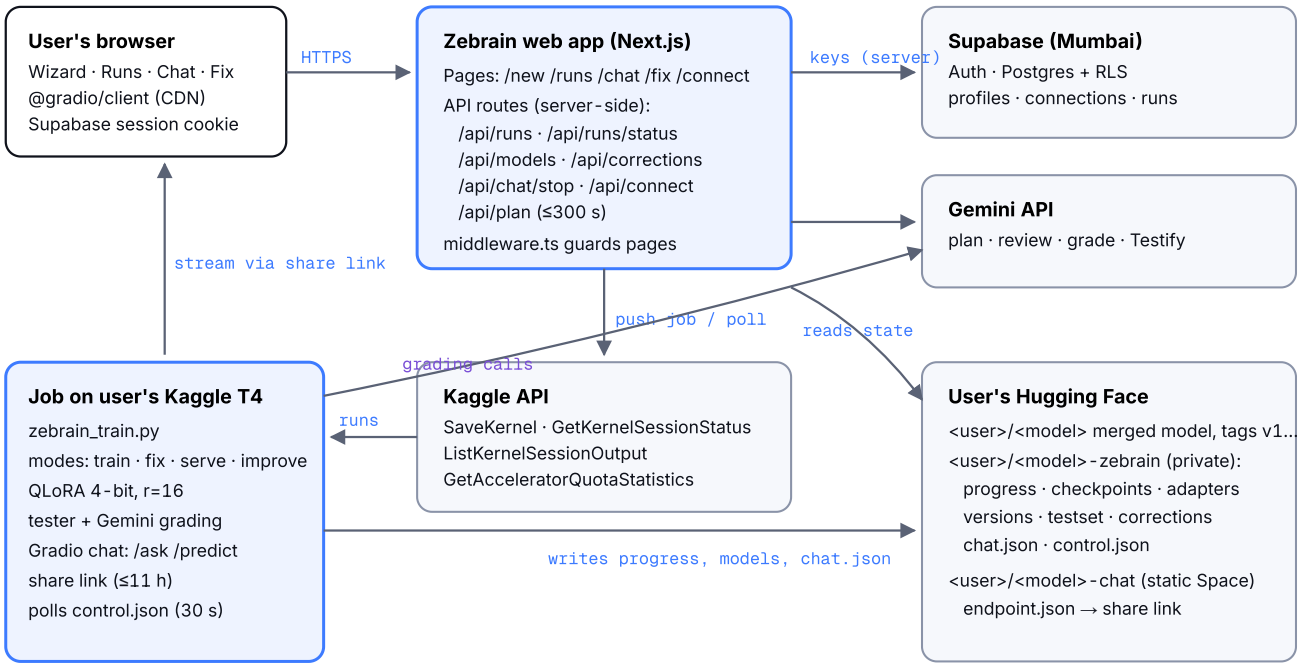


FIG 4.1 Zebrain as built. Blue boxes are Zebrain code. Grey boxes are external services the user or Zebrain holds accounts with. VERIFIED

The key architectural idea is that **Zebrain's servers do no GPU work**. They orchestrate: check accounts, push a self-contained Python job to the user's Kaggle, and read its progress from the user's Hugging Face state repo. The job writes everything it knows (progress, checkpoints, versions, test sets, the open chat's URL and API key) into that private repo, which makes runs resumable and the web app nearly stateless.

4.3

Frontend

- **Routes:** `/` (home, FAQ), `/pricing`, `/login`, `/connect` (Kaggle, HF, Gemini keys), `/new` (wizard), `/runs` (progress), `/chat` (dashboard, owner chat, API guide), `/fix` (corrections, identity, versions, Testify). VERIFIED
- **Live model access:** `lib/gradio.ts` loads the matching `@gradio/client` from a CDN at runtime (1.5.2 for Gradio 4/5, 2.7.1 for Gradio 6) and streams replies with `client.submit`. VERIFIED
- **State:** server data comes from API routes. There is no client-side store library. Runs are polled from `/api/runs/status`.
- RECOMMENDATION Pin the CDN client with Subresource Integrity, or bundle it, so a CDN compromise cannot run code on Zebrain pages.

4.4

Registration, authentication, accounts and permissions

Concern	Implementation
Sign-up / sign-in	Supabase Auth email + password, with email confirmation turned off in the dashboard. Google OAuth via `/auth/callback` once the provider is configured. VERIFIED
Session	Supabase SSR cookies. `middleware.ts` redirects signed-out visitors on `/new /runs /chat /fix /connect` to `/login?next=…`. VERIFIED
Authorisation	Row-level security on `profiles`, `connections` and `runs`: users only read and write their own rows. API routes resolve the user and their keys server-side via `userCreds()`. VERIFIED
Roles	One role (user) today. RECOMMENDATION Add `org` and `membership(role: owner|admin|member)` for Business clients and agencies, plus a separate staff/admin role behind SSO.
Account management	RECOMMENDATION Self-serve delete-account (cascades rows and revokes stored keys), export my data, change email, and forced re-auth for key changes.
Model API keys	Each chat session generates a random `zb_` + 32-hex key, shown only to the owner and checked by `/ask`. `/predict` is deliberately keyless, like a plain Gradio app. VERIFIED

4.5

Core business logic: the training job

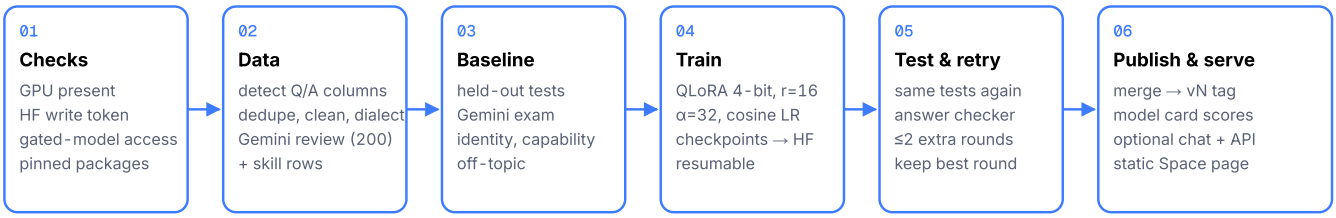


FIG 4.2 Train mode, as implemented in `lib/zebrain_train.py`. VERIFIED

Mode	What it does	Publish rule
train	The full pipeline above. Identity is built into the chat template (replacing Qwen's default system prompt), together with the answer-quality rules.	Best round by tester score
fix	Each correction becomes ~20 examples (reworded by Gemini if available), mixed with replayed old data. Continues from the latest adapter.	Only if the fix passes and old tests don't get worse
improve	"Testify": a Gemini exam → failures → weak topics → practice rows (never copies of test questions) → 2-epoch rounds with full skill replay.	Only if the best key score beats the starting version by more than 0.02
serve	Loads a version and opens the chat and API for 0.25–11 hours. Stops early when `control.json` names this run.	n/a

Pinned packages: transformers 4.46.3, peft 0.13.2, accelerate 1.1.1, datasets 3.1.0, huggingface_hub 0.26.5. Training uses batch 2 × gradient accumulation 8. Examples over 1,536 tokens are skipped rather than truncated. VERIFIED Known gap: retrain rounds in train mode don't replay skill rows, which caused identity drift in one internal run (100% → 71%). Improve mode fixes this, and train mode should adopt the same replay. RECOMMENDATION

4.6

Database schema, relationships and lifecycle

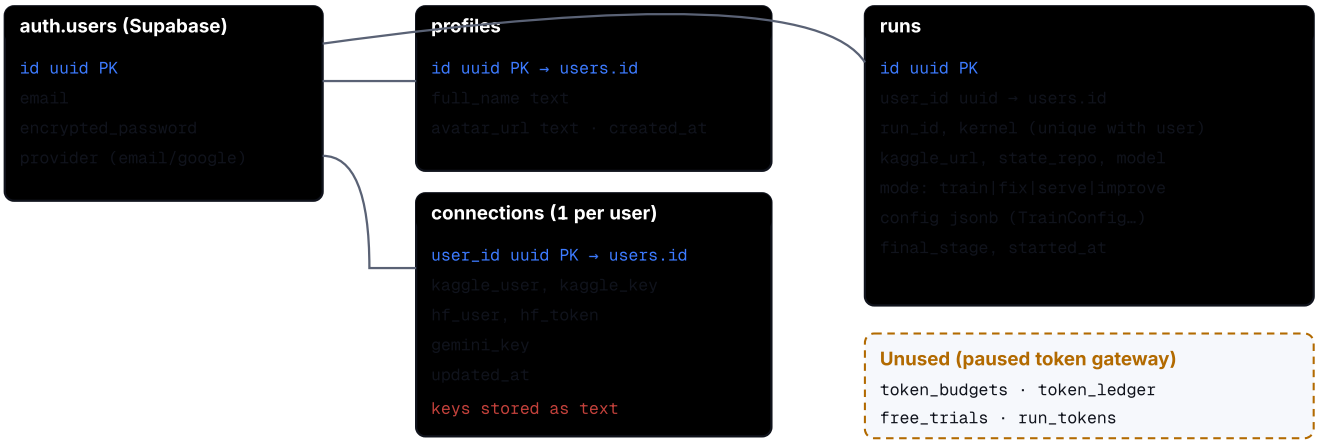


FIG 4.3 Public schema in Supabase as queried on 26 Oct 2026 [2]. One user → one profile, one connection row, many runs. VERIFIED

- **Indexes:** `runs` has a unique constraint on `(user_id, run_id, kernel)`, used by the upsert that resumes runs. RECOMMENDATION Add `runs(user_id, started_at desc)` for the runs list.
- **Unused tables:** four token-gateway tables and three functions (`reserve_tokens`, `settle_tokens`, `rebalance_tokens`) from a paused build. RECOMMENDATION Either finish the gateway or drop them, so the schema docu-

ments only what is real.

- **Proposed launch tables:** `subscriptions` (plan, status, period, gateway ids), `invoices`, `credits` (GPU-hours ledger), `hosted_models` (model, version, endpoint, plan), `api_keys` (hash, prefix, scopes, last_used), `usage_events` (model, messages, tokens, day, partitioned by month), `orgs`, `memberships`, `audit_log`.
- **Data lifecycle:** run rows are kept for the account's lifetime. Usage events are rolled up to daily after 90 days. Deleted accounts are purged within 30 days and their stored keys are deleted immediately (§4.12).

4.7

API design, conventions and error handling

Endpoint (all POST, JSON)	Request	Response
`/api/connect`	Kaggle user/key, HF token, Gemini key (blank keeps the saved value)	Checked accounts or `{error}`
`/api/plan`	`{job, mode?: "full"|"eval", datasets?}`	Chosen datasets with reasons, `eval_set`, Gemini model used
`/api/runs`	`{config: JobConfig, runId?}`	`{run}`: kernel, URLs, state repo
`/api/runs/status`	`{kernel, stateRepo}`	`{kaggle, progress, logTail}`
`/api/models`	`{stateRepo}`	Config, versions, pending corrections, open chat
`/api/corrections`	`{stateRepo, correction}`	`{ok, waiting}`
`/api/chat/stop`	`{stateRepo}`	`{ok}` or `{ok, alreadyOffline}`

```
# Start a run
POST /api/runs
{ "config": { "mode": "train",
    "model_name": "sql-assistant-1-5b",
    "base_model": "Qwen/Qwen2.5-Coder-1.5B-
Instruct",
    "datasets": [ ... ], "max_len": 1536,
    "chat_hours": 1 } }

200 { "run": { "runId": "0c58…", "kernel": "…",
        "stateRepo": "…-zebrain" } }
400 { "error": "Pick at least one dataset." }
```

```
# Model API on an open chat (streaming)
import { Client } from "@gradio/client";
const c = await Client.connect(SHARE_URL);
for await (const m of c.submit("/ask", {
  message: "Top 5 customers by revenue?",
  api_key: "zb_…", history: "[]" })) {
  if (m.type === "data") render(m.data[0]);
}
```

Conventions VERIFIED plus proposed additions

- Errors return `{error: "<plain sentence telling the user what to do>"}` with a 4xx/5xx status. Upstream failures (Kaggle, HF) return 502 with the provider's message. Messages are written for non-technical users ("Verify your phone number in Kaggle settings…").
- RECOMMENDATION For the public hosting API: version it (`/v1/chat`), adopt an OpenAI-compatible request shape so existing SDKs work, use `Authorization: Bearer zb_…`, idempotency keys on billable calls, rate-limit headers (`X-RateLimit-Remaining`), and documented error codes (`quota_exceeded`, `model_offline`, `invalid_key`).
- Validate every request body with a schema (e.g. Zod) at the route boundary.

4.8

Background jobs, serving and the proposed hosting platform

Today the "queue" is Kaggle itself: each run is a Kaggle script session, at most 2 GPU sessions per Kaggle account. The web app polls status. Turning a chat offline works by writing a signal into `control.json`, which the job checks every 30 s, because Kaggle offers no reliable external cancel. VERIFIED

Serving quality safeguards (built): replies stream token by token. Up to 1,536 new tokens per pass, and a reply that hits the cap continues automatically (up to 3 passes). A light repetition penalty (1.05) prevents loops. Before a reply is final, a checker looks for runaway encoded strings (e.g. data URLs, stopped as they start), unclosed code blocks, unbalanced brackets, Python syntax errors and over-length code when the user set a line limit. A failing answer is rewritten once, and is flagged if it still fails. VERIFIED

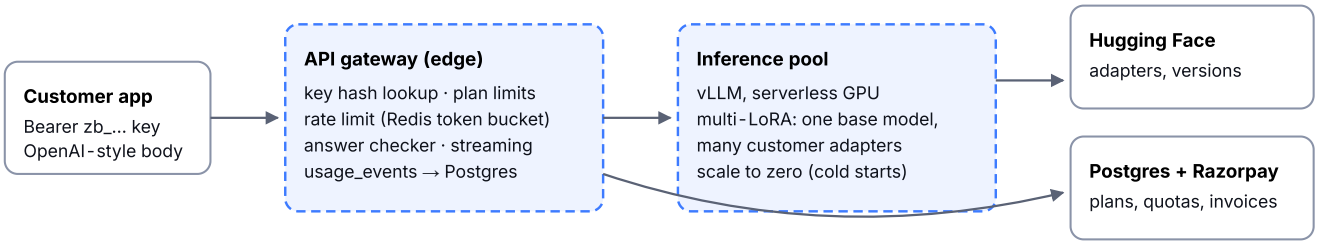


FIG 4.4 Proposed always-on hosting. Serving many small adapters on a shared base model (multi-LoRA) is the key cost lever for ≤3B models. RECOMMENDATION

Proposed scheduled tasks: nightly usage roll-up and quota reset at period start (cron); a daily check for stuck runs (Kaggle status stale over 2 h → notify the user); weekly key-age reminders; monthly invoice reconciliation against gateway webhooks.

4.9

Infrastructure, environments, CI/CD and observability

Environment	Purpose	Data
Local	Development (`npm run dev`)	Developer's own Supabase project and test Kaggle/HF accounts
Preview	One deployment per pull request	Staging Supabase project, seeded test users
Staging	Release candidate, full end-to-end run on a test Kaggle account	Staging project
Production	Customers	Production project, backups and PITR

- **Version control:** Git with trunk-based development and short-lived branches (§6.3). UNKNOWN The current working copy is not a Git repository. RECOMMENDATION Initialise a repo and push it to a private GitHub organisation this week.
- **CI on every PR:** `tsc --noEmit`, lint, `next build`, Python job tests (`tests/test_testify.py`, `tests/test_serving.py`, which exist today VERIFIED), and a dependency audit.
- **Weekly scheduled end-to-end:** a canary run trains a tiny model on a Zebrain-owned Kaggle account and checks every stage. This catches Kaggle image or package drift (the job pins package versions for this reason).
- **Observability:** error tracking (e.g. Sentry) on web and API, structured logs with request IDs, uptime checks on the site and gateway, and a funnel dashboard (Chapter 10). Run-level telemetry already exists in `progress.j-`

`son`. Aggregate its stage failures weekly.

4.10

Security: secrets, encryption, limits, backups and incidents

GAP: FIX BEFORE LAUNCH

Users' Kaggle keys, Hugging Face tokens and Gemini keys are stored as plain `text` columns in `connections` (protected by row-level security, but not encrypted by the application) [2]. In addition, the HF token and Gemini key are **embedded in each user's private Kaggle script** so the job can publish and grade [1]. A leaked script or database dump would expose working credentials.

Control	Recommendation	Priority
Encryption at rest	Encrypt key columns with a server-held key (Supabase Vault or envelope encryption via a KMS). Decrypt only inside API routes.	P0
Short-lived job credentials	Replace embedded tokens with a per-run token: the job exchanges it at a Zebrain endpoint for the minimum access it needs. It expires when the run ends. (This is what the paused `run_tokens` table was for.)	P0
Least privilege	Guide users to fine-grained HF tokens limited to their Zebrain repos. Show key age and a "rotate" prompt.	P1
Rate limiting	Per-IP and per-user limits on auth, `/api/plan` and `/api/runs`. Per-key limits on the hosting API (10 or 30 requests/min by plan).	P1
Headers and CSP	Content-Security-Policy allowing only the pinned client CDN, HSTS, frame-ancestors none.	P1
Backups and DR	Daily backups plus PITR. Quarterly restore drill. RPO 24 h / RTO 4 h for the web app. User models live on HF, outside Zebrain's blast radius.	P1
Secrets in code	Server secrets only in the hosting provider's environment. Secret scanning in CI. `.env.local` never committed.	P0
Incident response	Checklist in Appendix A15. Severity levels in §6.5. Breach notification duties under DPDP in §9.3.	P1

4.11

Performance, scalability, reliability and cost

- **Web tier** is stateless and scales horizontally. The heaviest route, `/api/plan`, is dominated by Gemini latency (probe 15 s, real calls up to 150 s). RECOMMENDATION Move it to a background job with progress events instead of one long request.
- **Training** scales with users for free, because each user brings their own Kaggle GPU. Paid training scales with the GPU budget, through a queue with per-user concurrency limits.
- **Chat latency:** streaming made the first words appear about 200 ms after submit in local tests (versus waiting for the whole reply before). VERIFIED (local Gradio 6 test)
- **Cost levers for hosting:** multi-LoRA on shared base models, 4-bit or 8-bit inference for ≤3B, scale-to-zero for Starter and warm pools for Pro, and response caching for repeated FAQ answers.

4.12

Data privacy, retention, deletion and access

Data	Where	Retention (proposed)
Account (email, name)	Supabase	Life of account + 30 days
Third-party keys	Supabase `connections`	Deleted immediately on disconnect or account deletion
Run metadata and config	Supabase `runs`	Life of account
Training data, test sets, corrections, chat state	User's own HF state repo	Controlled by the user
Chat messages	Not stored by Zebrain (the chat runs on the user's GPU). Corrections are saved only when the user chooses.	n/a VERIFIED
Hosting usage (future)	Postgres `usage_events`	Counts kept 24 months. Message content not logged by default.

Access: staff access to production data needs SSO, MFA and a ticket reason, and every access is logged. **Third-party processing:** prompts sent to Gemini on its free tier may be used by Google to improve its products, per Google's terms [9]. Disclose this in the privacy policy, and use a paid tier for Business clients' data.

4.13

AI/LLM infrastructure: models, prompts, evaluations, safeguards

Component	Current design VERIFIED	Next step RECOMMENDATION
Base models	Small open instruct models (e.g. Qwen2.5-Coder-1.5B). Gated models checked for licence acceptance first. Llama naming rules enforced.	Curated catalogue with a measured quality/latency score per size.
Gemini selection	Ranks `gemini-X.Y-flash` models and probes each with a 15 s request, because newest models were sometimes overloaded or silent. Falls back to the next. Calls spaced 6.5 s apart in jobs for free-tier limits.	Provider abstraction, budget caps, response caching.
Prompts	Inline in code (planner, reviewer, grader, Testify, paraphrase).	Prompt registry: versioned prompts in files, with golden tests per prompt.
Evaluations	Held-out rows, Gemini-written exam, identity, capability and off-topic checks. Gemini grading with a rules fallback. The answer checker counts broken code as wrong.	Persist eval results per version in Postgres. Publish a public benchmark per use case.
Safeguards	Off-topic refusal training. Quality rules in the system prompt. Testify never trains on test-question copies (exact or token-F1 > 0.8).	Content-safety filter on hosting outputs. Abuse reporting. Block models trained to impersonate real people or brands without rights.

05

Frontend, UI/UX & Design System

The visual and interaction system Zebrain already ships (black, one blue, Inter and a pixel wordmark) codified into tokens, components and rules.

5.1 Brand identity and principles

5.2 Colour and type tokens

5.3 Layout, spacing and responsiveness

5.4 Components

5.5 Information architecture and journeys

5.6 Key screens

5.7 States, accessibility and handoff

5.1

Brand identity and design principles

Zebrain's interface uses a black page, a single brand blue, the Inter typeface, a pixel font (Silkscreen) for the wordmark and accents, a blue "light beam" gradient in the hero, pill-shaped buttons and chat bubbles styled like Gradio's. VERIFIED (`tailwind.config.ts` , `globals.css` , `layout.tsx`) No logo file or brand guideline document exists. UNKNOWN This chapter codifies what the product already does. Anything new is marked as a recommendation.

ONE COLOUR

Blue `#3d7bff` is reserved for action and focus: buttons, focused links, the beam. Never decoration.

PLAIN WORDS

Labels say what happens ("Train on Kaggle", "Turn offline"). No AI jargon in primary UI.

SHOW THE PROOF

Scores, before/after answers and logs are first-class UI, because trust comes from evidence.

Logo and wordmark usage

RECOMMENDATION Until a designed logo exists, the wordmark is the word ZEBRAIN set in Silkscreen, lower-case, in paper white on black or black on white. Keep clear space of at least the height of the "z" on all sides. Do not stretch it, outline it, add gradients to the letters, or set it in any other font. Commission a proper logo (mark + wordmark) before paid marketing begins, and check trademark availability first (§9.2).

5.2

Colour and typography tokens



FIG 5.1 Colour tokens from `tailwind.config.ts` . VERIFIED

Token	Family	Size / weight	Use
`display`	Inter	40–64 px / 600, tracking −2%	Hero headlines
`h1-h3`	Inter	28 / 22 / 18 px, 600	Page and section titles
`body`	Inter	15–16 px / 400, line-height 1.55	Copy, forms
`mono`	Geist Mono	13 px / 400–500	Model replies, code, numbers, logs
`pixel`	Silkscreen	Wordmark, eyebrow labels	Brand moments only. Never body text.

Contrast: paper on ink is about 19:1 and mute on ink about 6.8:1 (both pass WCAG AA). Accent on ink is about 5.1:1, which passes AA for normal text. White on accent is about 4.1:1: use it for buttons with 15 px or larger semibold labels (AA large), or darken the button shade if smaller text is needed. RECOMMENDATION Verify with a contrast checker when tokens change.

5.3

Layout, grids, spacing and responsive behaviour

Rule	Value
Spacing scale	4 px base: 4, 8, 12, 16, 24, 32, 48, 64, 96 (Tailwind 1–24)
Content width	Reading pages max 768 px (`max-w-3xl`). Dashboards max 1152 px.
Grid	12 columns, 24 px gutter on desktop · 8 columns on tablet · 4 columns with a 16 px side gutter on phones
Breakpoints	sm 640 · md 768 · lg 1024 · xl 1280 (Tailwind defaults)
Radius	Pills (buttons, chips) 999 px · cards 12 px · inputs 8 px · chat bubbles 16 px with one 6 px corner
Responsive rules	Tables scroll inside their container or hide secondary columns (as the pricing table does). The chat input is sticky at the bottom. No horizontal page scroll at 360 px.

5.4

Component library

Component	Spec	Do / Don't
Button (primary)	Pill, accent fill, white semibold label, 40 px tall, 2 px focus ring in paper	One primary per view. Disabled at 35% opacity with the reason shown nearby.
Button (link)	Underlined on hover, mute → paper	For secondary actions ("Cancel", "Show").
Input / textarea	Panel fill, line border, focus border paper, label above	Never use a placeholder as the label.
Chip	Pill outline, `aria-pressed` for toggles (model size picker)	Use for choices of 2–7 options.
Card	Panel fill, 1 px line, 12 px radius	No drop shadows on black. Hierarchy comes from spacing.
Table	Hairline rows, mute header, right-aligned numbers in mono	Zebra stripes only on very wide tables.
Chat bubble	User: accent fill, right. Model: panel fill with line border, mono text, left.	"Wrong answer?" link under each finished model reply.
Stage list	Ordered stages with state (waiting, running, done, failed) and a one-line message	Always show what's happening and roughly how long it takes.
Modal / confirm	Used only for destructive or costly actions ("Turn offline?")	Primary button names the action, not "OK".
Toast / status	`role="status"` line near the action	Errors stay until fixed. Successes fade after 4 s.
Empty state	One sentence + one primary action ("No models yet. Describe a job to train your first.")	No illustrations needed.

5.5

Information architecture and user journeys

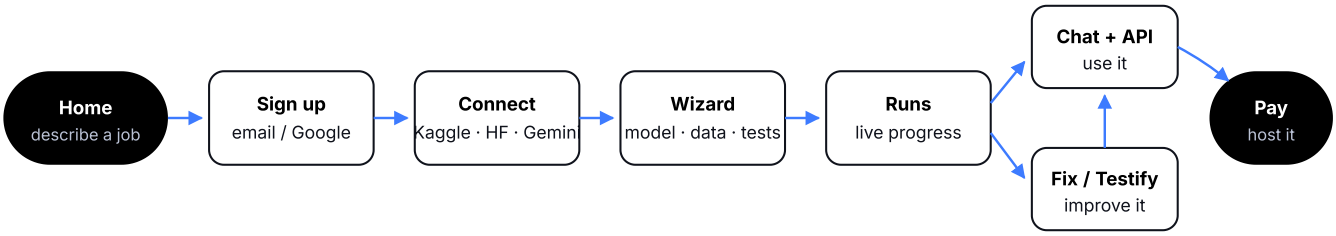


FIG 5.2 Core journey. Connect is the biggest drop-off risk (Kaggle phone verification). The "Pay" step is proposed. RECOMMENDATION

Onboarding fixes to prioritise: (1) a Connect checklist that validates each key live and explains Kaggle phone verification with screenshots; (2) a "try a ready-made model" chat before connecting anything, so value shows up in under a minute; (3) remember the job description typed on the home page through sign-up. RECOMMENDATION

5.6

Key screens

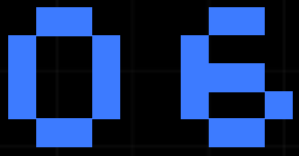
Screen	Purpose and content	Status
Landing	Beam hero with "describe a job" input, how it works, comparison with ChatGPT/Gemini, FAQ	BUILT
Auth	Email/password and Google, with a `next` redirect back to the intended page	BUILT
Connect	Three key fields with live checks. A blank field keeps the saved key.	BUILT
Wizard	Job → base model → data ("Let Gemini choose" or search) → test set → identity → chat hours → Train	BUILT
Runs	Stage list, loss, scores table (test, exam, identity, capability, off-topic), log tail, resume, import of older runs	BUILT
Chat dashboard	Status, Turn offline, API endpoint, Hugging Face link, owner chat (streaming), API guide with JS/Python snippets	BUILT
Fix	Corrections queue, identity editor, versions, Testify with Gemini (1–3 rounds)	BUILT
Billing / settings	Plan, usage meter, invoices, API keys, delete account	PROPOSED
Admin	Users, runs by failure stage, credits, abuse reports, feature flags (staff only, SSO)	PROPOSED

5.7

States, accessibility and design handoff

- **Loading:** say what is happening ("Gemini is reading sample rows... up to 2 minutes"). Never use a spinner without words for waits over 2 s.
- **Errors:** say what went wrong and what to do, in one sentence, next to the cause. API errors already follow this pattern. VERIFIED
- **Success:** state the result and the next step ("Saved. 3 corrections waiting on the Fix page").
- **Edge cases:** chat closed mid-reply, Kaggle quota exhausted, gated model not accepted, Gemini overloaded, HF read-only token. Each gets a specific message.

- **Accessibility:** keyboard reachable everything, visible focus rings, `aria-live` on streaming chat and status, labelled inputs, `prefers-reduced-motion` respected, targets of 40 px or more on touch.
- **Handoff:** tokens live in `tailwind.config.ts`, which is the single source. Design files mirror the token names exactly. Each component has a short doc (purpose, props, states, accessibility notes) next to its code. RECOMMENDATION
- **Frontend standards:** server components by default, client components only for interactivity, no new UI library without a decision record, and every page must pass a 360 px width check and keyboard-only use before merge.



Product Development & Engineering Workflow

How Zebrain decides what to build, specifies it, ships it safely and learns from users.

6.1 Roadmap and prioritisation

6.2 Requirements and cadence

6.3 Branching, reviews and releases

6.4 Testing strategy

6.5 Bugs, severity and triage

6.6 Tech debt and documentation

6.7 Feedback, analytics and experiments

6.1

Product roadmap and prioritisation

Window	Theme	Items (in priority order)
Now (0–6 wks)	Launch safely	Git repo + CI · production deploy · key encryption and per-run tokens · Kaggle terms opinion · skill replay in train-mode retrain rounds · booking page for Business
Next (6–12 wks)	Get paid	Razorpay (plans, top-ups, credits) · hosting ≤3B (multi-LoRA) · usage metering · billing page · onboarding checklist
Then (3–6 mo)	Better models	Dataset generation (~2,000 examples per job, LLM-written with web search) · document upload → topics → examples · paid L40S training · eval history per version
Later (6–12 mo)	Scale	Orgs and teams · agency programme · website chat widget · Indian-language support · H100 training · OpenAI-compatible API

Why data generation ranks above more training rounds: in the internal coding-model run, extra rounds moved the held-out test score only from 38% to 42% and plateaued, while 25-question exams can't cover a domain. Small models are data-limited. Producing about 2,000 high-quality, task-specific examples per job is the biggest quality lever. VERIFIED (internal run) + RECOMMENDATION

6.2

Requirements, user stories and development cadence

- **Methodology:** two-week cycles (lightweight Scrum), plus a continuous-flow lane for bugs. Justification: one to four engineers, fast-changing external APIs, and a need for a weekly demo on real models.
- **Every feature starts with a 1–2 page PRD** (Appendix A5): problem, users, success metric, scope / out of scope, user stories, acceptance criteria, risks, rollout.
- **User story format:** "As a *shop owner*, I want *to fix one wrong answer from the chat*, so that *the next version answers correctly*."
- **Acceptance criteria in Given/When/Then.** Example: Given a saved correction, when I start a fix run, then the published version answers the corrected question correctly *and* no earlier test score drops by more than 2 points.

6.3

Environments, branching, reviews and release management

- **Branching:** trunk-based. Short branches `feat/…`, `fix/…`, merged by squash after review. `main` is always deployable.
- **Pull requests:** under 400 lines where possible. The description covers what, why, screenshots and test evidence. One approving review, two for auth, keys, billing or the training job's publish logic.
- **Releases:** continuous deployment of the web app from `main` behind feature flags. The training job is versioned (`JOB_VERSION`), and each run records which version it used, so a regression can be traced to a job release.

Release checklist

- [ ] CI green: typecheck, lint, build, Python job tests
- [ ] Staging canary run completed all stages on a test Kaggle account (for job changes)
- [ ] Database migrations reviewed, reversible, and applied to staging first
- [ ] Feature flag default set. Rollback plan written in the PR.
- [ ] Changelog entry and docs/API guide updated
- [ ] Error-rate and run-failure dashboards watched for 30 minutes after deploy

6.4

Testing strategy

Level	What	Today
Unit	Pure functions: pricing scale, model ranking, answer checker, practice-row de-duplication, encoders	EXISTS (PYTHON)
Integration	Gradio endpoints with a fake model, API routes with mocked Kaggle/HF/Gemini	EXISTS (SERVING)
End-to-end	Browser test: sign up → connect (test keys) → wizard → run starts. Weekly real canary run.	PROPOSED
Model quality	Golden prompts per base model and use case. Scores must not regress between job versions.	PROPOSED
Performance	Time to first token and tokens/s on T4 for 1.5B and 3B. Hosting load test before launch.	PROPOSED
Security	Dependency audit in CI, RLS policy tests (user A cannot read user B), secret scanning, annual external pen test	PROPOSED

6.5

Bug reporting, severity and triage

Sev	Definition	Example	Response / fix
S1	Security exposure, data loss, or all runs/hosting down	Keys exposed. Sign-in broken for everyone.	15 min / same day
S2	Core flow broken for many users, no workaround	Kaggle image change breaks the job install	2 h / 2 days
S3	Feature broken with a workaround, or one user's flow	Testify fails without a Gemini key on one path	1 day / next cycle
S4	Cosmetic or minor	Misaligned table on mobile	Backlog

Bug report template: Appendix A5 (bug section). Triage happens daily: the CTO sets severity and owner, and S1/S2 open an incident (Appendix A15).

6.6

Technical debt and documentation standards

- Keep a debt register in the tracker, tagged `debt`, and spend 15–20% of each cycle on it. Current known items: unused token tables, train-mode retrain skill replay, the CDN client without SRI, plaintext key columns, `/api/plan` as a long request, booking-link placeholder, no Git history.
- Documentation: README for setup, this manual for operations, decision records for one-way doors, and doc-strings that explain *why* (the codebase already does this well). VERIFIED

6.7

User feedback, analytics and experimentation

- **In-product feedback:** the "Wrong answer?" and correction flow is also a feedback channel. Add a 1-click "Was this run useful?" after each publish.
- **Analytics events** (privacy-friendly, no message content): `signup`, `connect_completed{service}`, `plan_requested`, `run_started{mode}`, `run_finished{stage, scores}`, `chat_opened`, `api_called`, `correction_saved`, `checkout_started`, `subscribed{plan}`.
- **Experiments:** one at a time on the onboarding funnel. Write the hypothesis and the minimum detectable effect first, and run each test for at least 2 weeks or until it reaches its planned sample size.

07

Marketing, Branding & Growth

How Zebrain gets found, gets tried and gets talked about, with a 90-day plan that has owners, budgets and targets.

7.1 Positioning and messaging

7.2 Audiences and segments

7.3 The funnel

7.4 Channels

7.5 Content pillars and campaigns

7.6 Budget and KPIs

7.7 90-day execution plan

7.1

Positioning, messaging and tone of voice

RECOMMENDATION

One-line message: "Make an AI model of your own. Free to train, yours to keep."
Proof points: trains on your own free GPU · tested before and after · answers in your name and stays on topic · lives in your Hugging Face account.

Tone rule	Write this	Not this
Plain words	"Teach it your answers"	"Fine-tune with QLoRA adapters"
Concrete	"Can I return shoes after 20 days? → Yes, you have 30 days..."	"Unlock AI-powered customer delight"
Honest	"Small models are great at one job, not at everything"	"As smart as GPT"
Confident, calm	"Your model, tested."	"Revolutionary!!"

The pricing page already uses this voice: the shoe-shop before/after example and the honest per-message comparison with ChatGPT and Gemini. VERIFIED

7.2

Ideal customer profiles and segmentation

Segment	ICP signal	Entry offer	Priority
Indian D2C and SMB support teams	Shopify/WooCommerce store, FAQ page, WhatsApp support, 2–50 staff	Business plan pilot	1
Indie developers and students	Active on GitHub, Kaggle or Hugging Face. Builds side projects.	Free training → Starter	2
Agencies and freelancers	Build websites/chatbots for clients	Reseller margin on Business	3
Educators and coaching centres	Course material, repeated student questions	Free → Starter	4

7.3

Marketing funnel (AAARRR)

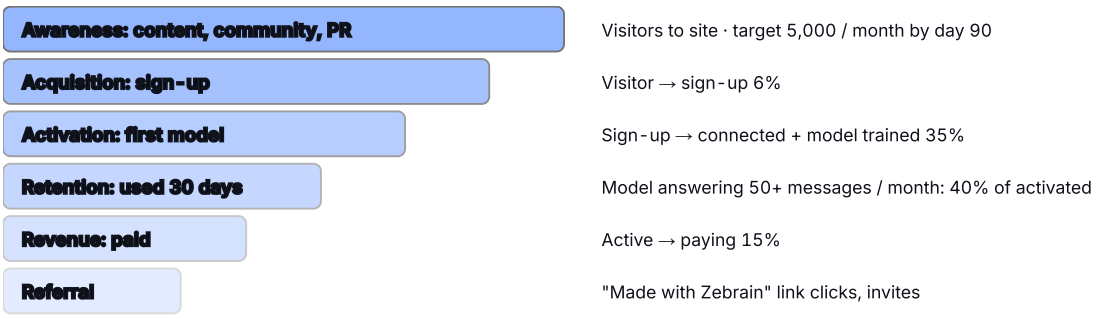


FIG 7.1 Funnel with target conversion rates. ILLUSTRATIVE TARGETS Replace them with measured baselines after 30 days.

Built-in referral loop: free-tier models carry a small "Made with Zebrain" identity, and every model card and public chat page links back. VERIFIED (pricing copy, static page footer) Every free model is an advertisement.

7.4

Channels

Channel	Plan	Cost
Website and SEO	Pages for each use case ("make a SQL model", "customer support bot for Shopify"). Guides targeting "fine-tune Qwen free", "train LLM on Kaggle", "custom chatbot India".	Time
Hugging Face and Kaggle	Public showcase models with model cards. A Kaggle notebook explaining the method. Answer community questions.	Time
YouTube and shorts	"From sentence to my own AI in 30 minutes" screen recordings. Before/after answer clips.	Low
LinkedIn / X	Founder build-in-public posts with real test scores and lessons.	Time
Communities	Indian developer communities, college tech clubs, Shopify seller groups (follow each group's rules).	Time
Email	Onboarding sequence (5 emails) and a monthly "what models people made" digest.	Low
PR	Launch on Product Hunt and Hacker News ("Show HN"). Pitch Indian startup media with a data story.	Low
Paid	Only after activation is above 30%: Google Search on high-intent terms, Meta retargeting of site visitors.	₹30–40k / month from month 3
Partnerships	Agencies (reseller), colleges (workshops), Shopify app developers.	Rev-share

7.5

Content pillars, example posts and campaigns

Pillar	Example
Proof	"We trained a 1.5B SQL model for ₹0. Exam score went from 73% to 78%. Here's what moved it."
How-to	"Turn your FAQ page into a support bot in 4 steps."
Ownership	"Why your support bot shouldn't live on someone else's server."
Behind the build	"How we stopped our model from saying 'I am Qwen'."

Campaign concepts

- **"My First Model" week:** live workshop + template + leaderboard of the best-tested community models. CTA: "Train yours free."
- **"Fix it in one click":** show a wrong answer, a correction, and the fixed version. CTA: "Try the Fix page."
- **Business pilot offer:** "First 5 stores get a done-for-you support bot at the pilot price." CTA: "Book a 20-minute call."

7.6

Budget allocation and KPIs

90-DAY MARKETING BUDGET ILLUSTRATIVE · TOTAL ₹1,20,000

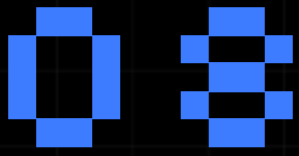
Line	₹	Share
Video editing and design freelancers	30,000	25%
Paid search + retargeting (month 3 only)	35,000	29%
Workshops and events	20,000	17%
Tools (email, analytics, scheduling)	15,000	12%
Launch assets and contingency	20,000	17%

KPIs: unique visitors, visitor→sign-up rate, activation rate (first model trained within 7 days), cost per activated user, organic share of sign-ups, email open/click rates, and Business calls booked. Definitions are in Chapter 10.

7.7

90-day marketing execution plan

Weeks	Deliverables	Owner	Budget	Success metric
1–2	Message test: 10 customer interviews. Final positioning. Analytics events live.	CEO	₹0	Top 3 pains named by 6 or more of 10
3–4	3 use-case landing pages. 3 showcase models on HF. Onboarding emails.	Growth	₹15k	Pages indexed. 3 public models with scores.
5–6	"My First Model" workshop (online). 4 short videos. Kaggle explainer notebook.	Growth	₹25k	150 sign-ups. 40 trained models.
7–8	Product Hunt + Show HN launch. Founder posts 3×/week. Agency outreach (20).	CEO	₹20k	1,000 visitors on launch day. 3 agency calls.
9–10	Case study from the first Business pilot. Comparison page vs. generic chatbots.	Success	₹10k	1 published case study
11–13	Paid search test on 20 high-intent keywords. Retargeting. Review and decide.	Growth	₹50k	Cost per activated user under ₹400, else stop paid



Sales, Customer Acquisition & Revenue

A repeatable sales system: self-serve for makers, consultative for Business clients.

8.1 Sales motions

8.2 Discovery and qualification

8.3 CRM and pipeline

8.4 Outreach and scripts

8.5 Demos, proposals, closing

8.6 Onboarding, success and retention

8.7 Forecasts and targets

8.1

Sales motions

SELF-SERVE (FREE → STARTER/PRO)

No salesperson involved. The product, onboarding emails and in-app prompts do the selling, and the trigger is usage: "Your free chat closes in 1 hour. Keep it always on for ₹499/month."

CONSULTATIVE (BUSINESS PLAN)

Founder-led and later Success-led. A discovery call, then a pilot SOW, then a model passing the client's own acceptance questions, then go-live. Price is agreed per client, as the pricing page states. VERIFIED

8.2

Lead generation, discovery and qualification

Lead sources: "Book a meeting" on the pricing page (link still a placeholder VERIFIED), inbound from content, agency referrals, and outbound to stores with public FAQ pages.

QUALIFICATION (ADAPTED BANT)

Question	Qualified if
Need: how many repetitive questions per week?	More than 100, or a clearly painful repeated task
Data: do you have FAQs, policies or past chats?	Yes, or willing to write 30+ Q&A pairs
Authority: who decides?	Owner or ops head on the call
Budget: what do you spend on support today?	Can pay the pilot price
Timing: when should it be live?	Within 60 days

Discovery call script (20 minutes) ILLUSTRATIVE

1. **Context (3 min):** "Tell me how customers reach you today, and what they ask most."
2. **Pain (5 min):** "What happens when nobody answers quickly? Have you tried a chatbot? What went wrong?"
3. **Data (4 min):** "Where do the right answers live today: FAQ page, a doc, someone's head?"
4. **Show (5 min):** the before/after demo on a sample store model, including an off-topic refusal and a correction.
5. **Next step (3 min):** "If I send a pilot plan with 25 of your real questions as the test, could we decide by Friday?"

8.3

CRM setup and pipeline stages

Stage	Exit criteria	Win prob.
1 Lead	Contact info + source captured	5%
2 Qualified	Passes 4 of 5 qualification questions	15%
3 Discovery done	Pain, data and decision-maker documented	30%
4 Pilot proposed	SOW with acceptance questions sent	50%
5 Pilot running	SOW signed, data received	70%
6 Won	Model passes acceptance, invoice paid	100%
Lost / Nurture	Reason recorded (price, timing, quality, no data)	0%

Use one CRM (HubSpot free tier or a simple Notion/Sheets pipeline at the start). RECOMMENDATION Required fields: source, segment, stage, next step with date, value, lost reason.

8.4

Outreach strategy and email sequence

Outbound sequence for stores ILLUSTRATIVE

Day 1 · Subject: "Your return-policy questions, answered at 2 a.m."
Hi {Name}, I noticed {Store}'s FAQ covers {topic}. We build small AI models that answer those questions in your words and stay on topic. Could I show you one trained on your FAQ, at no cost, in a 15-minute call?

Day 4: a 40-second video of a model answering 3 of their real FAQ questions.
Day 9: case study or test scores from a similar store.
Day 16: "Should I close the loop?" break-up email.

Comply with anti-spam norms: identify the sender, include an opt-out, and only email business addresses published for contact.

8.5

Demos, proposals, negotiation and closing

- **Demo rule:** always demo on the prospect's own questions (3–5 from their FAQ), with a model trained before the call, never generic examples.
- **Proposal:** 2 pages. Problem, pilot scope, the acceptance test (their 25 questions, pass mark agreed upfront), timeline, price, what happens after the pilot.
- **Negotiation:** trade scope, not price. A smaller model, fewer languages or a longer timeline lowers the price, while discounts are given only for annual prepayment.
- **Close:** the pilot converts to a monthly plan automatically unless the client cancels within 14 days of go-live.

8.6

Onboarding, support, retention and escalation

Moment	Action	Owner
Day 0	Welcome call, shared channel, data checklist	Success
Week 1	First model plus test report shared	Success
Week 2	Client corrections → fix run → Testify	Success
Go-live	Chat widget live. Usage dashboard shared.	Success + CTO
Monthly	Review wrong answers, retrain if needed, report	Success
Churn signals	Usage down 40% month on month, unanswered corrections, late invoice → call within 48 h	Success

Support SLAs (Appendix A14): first response within 4 business hours on Business and 1 business day on Starter/Pro. Escalation path: Success → CTO for model or platform issues → CEO for commercial disputes.

8.7

Sales forecasts, quotas and pipeline reporting

BUSINESS-PLAN PIPELINE TARGETS ILLUSTRATIVE

Quarter	Qualified leads	Pilots	Won	Quota (new MRR)
Q1	20	4	1	₹25,000
Q2	30	6	1	₹40,000
Q3	40	6	1	₹40,000
Q4	50	8	2	₹80,000

```
Weighted pipeline = Σ (deal value × stage win probability)  ·  Coverage = weighted pipeline ÷ quarter quota (target ≥ 3×)
```

09

Legal, Security, Risk & Compliance

The legal and operational obligations Zebrain must meet, and the places where a qualified professional must decide.

9.1 Formation and governance

9.2 IP, trademarks and licences

9.3 Data protection

9.4 Platform terms

9.5 Customer and people agreements

9.6 Risk register and continuity

NOT LEGAL ADVICE

This chapter identifies issues and proposes an approach. Every item marked "counsel" needs a qualified Indian lawyer, CA or company secretary before action.

9.1

Company formation and governance

- **Entity:** a private limited company incorporated through the MCA's SPICe+ process is the usual vehicle for a venture-funded Indian startup [17]. ASSUMPTION Then apply for DPIIT startup recognition.
- **Founders' agreement:** roles, vesting (4 years with a 1-year cliff), IP assignment, decision rights, and what happens if a founder leaves.
- **Governance:** board meetings as required by law, minutes, a statutory register, and a decision log for one-way doors (§2.3).

9.2

Intellectual property, trademarks and software licences

Asset	Position and action
Zebrain code	Owned by the company through IP assignments (§2.7). Keep it private. Consider open-sourcing the job script later as a marketing asset.
Name "Zebrain"	Run a trademark search in classes 9 and 42 (software and SaaS) in India, plus the main export markets, before brand spend. Other AI businesses use similar names. COUNSEL
Users' models	Belong to users: they live in users' accounts and are derived from their data and open base models. Say so in the terms.
Base models	Each base model carries its own licence. Apache-2.0 models (e.g. Qwen2.5-Coder-1.5B-Instruct [11]) are permissive. Llama derivatives must follow Meta's licence, including "Built with Llama" naming, which is implemented [10]. Gemma has its own terms.
Datasets	The planner prefers open-licensed datasets. VERIFIED Record each dataset's licence on the model card. Block datasets whose licence forbids commercial use when the user's plan is commercial.
Open-source dependencies	Keep a licence inventory (Next.js, Supabase clients, transformers, PEFT, Gradio). All are permissive today.

9.3

Data protection and cybersecurity responsibilities

India's Digital Personal Data Protection (DPDP) Rules were notified on 13 November 2025. They phase in over 18 months, with most substantive obligations (notice, consent, data-principal rights) due from May 2027 [4]. Zebrain is a *data fiduciary* for account data and should prepare now:

- [ ] Privacy notice in plain language: what is collected (account, keys, run metadata), why, retention (§4.12), processors (Supabase, Gemini, Kaggle, Hugging Face, payments).
- [ ] Consent and withdrawal: sign-up consent, and delete-account that really deletes.
- [ ] Data-principal rights process: access, correction and erasure requests answered within the statutory period.

- [ ] Breach response: notify the Data Protection Board and affected users as the Rules require (Appendix A15). COUNSEL on timelines.
- [ ] Processor contracts (DPAs) with every vendor that touches user data.
- [ ] Business clients: Zebrain is usually a *processor* for their end-customers' chats. The MSA must allocate responsibilities.
- [ ] Security baseline: Chapter 4.10 controls, MFA for staff, least privilege, annual penetration test.

9.4

Platform terms: Kaggle, Hugging Face, Google

HIGHEST-PRIORITY LEGAL QUESTION

Zebrain's free tier runs a commercial service's job on users' own Kaggle accounts. Kaggle's terms of use govern how its free compute may be used [15]. Before launch, counsel should review whether this pattern is permitted, what restrictions apply (e.g. to long-running chat sessions), and how Zebrain should describe it to users. Ask Kaggle directly if the opinion is unclear.

- **Hugging Face:** users publish to their own accounts with their own tokens. Zebrain uses free static Spaces, because Gradio/Docker Spaces need PRO. Review HF's terms for automated repo creation at scale.
- **Gemini API:** follow the additional terms. On unpaid tiers, submitted content may be used to improve Google products [9], so Business data should go through a paid tier.

9.5

Customer, vendor, employee and contractor agreements

Document	Must cover
Terms of Service	Acceptable use (no illegal, deceptive or impersonating models), ownership of models, third-party accounts are the user's responsibility, no warranty of model accuracy, liability cap, plan limits, termination
Privacy Policy	DPDP-aligned notice (§9.3)
Business MSA + SOW	Scope, acceptance test, fees, payment terms, data processing, confidentiality, IP, service levels, termination and data return
Refund / credit policy	GPU-hour credits for failed paid runs (as promised on the pricing page), subscription cancellation
Employment / contractor	IP assignment, confidentiality, notice periods, statutory benefits
NDA	Mutual, for Business discovery where clients share documents

9.6

Risk register, incident management and business continuity

The full register template is Appendix A19. The top risks are scored below (likelihood × impact, each from 1 to 5).

Risk	L	I	Score	Mitigation	Owner
Kaggle terms or quota block the free tier	3	5	15	Legal opinion, paid GPU path, exportable job	CEO
Stored credentials leak	2	5	10	Encryption, per-run tokens, incident plan	CTO
Model quality disappoints paying clients	4	4	16	Acceptance tests in SOW, larger models, data generation	CTO
Gemini outage or terms change	3	3	9	Rules fallback (exists), second provider	CTO
Cash runs out before funding	3	5	15	Month-5 decision rule (§3.8)	CEO
A user's model produces harmful output	2	4	8	Acceptable-use policy, safety filter on hosting, takedown process	CEO
Key-person dependency (single founder)	4	4	16	Documentation (this manual), first engineering hire, shared credentials vault	CEO

Business continuity: because models, adapters and run state live in users' Hugging Face accounts, a Zebrain outage stops new runs but loses no user models. Keep infrastructure as code, keep a restore runbook, and keep a second admin with vault access.

10

Analytics & Performance Management

One measurement framework for product, growth, sales, customers, engineering and money, with an owner and a decision for every number.

10.1 KPI catalogue

10.2 Dashboards and cadence

10.1

KPI catalogue

KPI	Definition and calculation	Source	Owner	Review	Decision it drives
Models in active use (north star)	Models with 50 or more messages in the last 30 days (Kaggle chat + hosting)	usage_events, chat telemetry	CEO	Weekly	Overall product health. Where to focus.
Activation rate	Sign-ups who complete a training run within 7 days ÷ sign-ups	Analytics events	Product	Weekly	Onboarding fixes
Connect drop-off	Sign-ups that never save all required keys ÷ sign-ups, by service	connections table	Product	Weekly	Kaggle/HF guidance work
Run success rate	Runs reaching "published" ÷ runs started, split by failure stage and cause	runs + progress.json	CTO	Weekly	Reliability backlog
Model quality lift	Median (after − before) test score per published version. Exam pass rate.	Run reports	CTO	Monthly	Data and training investments
Visitor → sign-up	Sign-ups ÷ unique visitors	Web analytics	Growth	Weekly	Landing pages, messaging
CAC	Sales + marketing spend ÷ new paying customers	Books + billing	Growth	Monthly	Channel budgets
Attributed sign-ups	Sign-ups by first-touch UTM source	Analytics	Growth	Weekly	Channel mix
MRR / net new MRR	Sum of active monthly plan values. New + expansion − contraction − churn.	Billing	CEO	Weekly	Forecast, hiring
Logo churn	Paying customers lost in month ÷ paying customers at start of month	Billing	Success	Monthly	Retention programmes
Pipeline coverage	Weighted pipeline ÷ quarter quota	CRM	CEO	Weekly	Outbound effort
CSAT / NPS	Post-support rating (1–5). Quarterly NPS survey.	Support tool, survey	Success	Monthly	Support process, roadmap
Uptime and p95 latency	Successful requests ÷ total. p95 time to first token on hosting.	Monitoring	CTO	Weekly	Infra spend, capacity
Deploy frequency, change failure rate	Deploys per week. Share of deploys causing an incident or rollback.	CI/CD	CTO	Monthly	Engineering process
Gross margin	(Revenue − cost of revenue) ÷ revenue	Books	Finance	Monthly	Pricing, GPU provider choice
Net burn and runway	Cash out − cash in. Cash ÷ 3-month average burn.	Bank + books	Finance	Monthly	Spend freeze, fundraising timing

10.2

Executive dashboards and reporting cadence

Report	Audience	Cadence	Contents
Weekly executive report (A17)	Team	Monday	North star, funnel, runs, MRR, top 3 risks, top 3 priorities
Monthly business review	Team + advisors	Monthly	P&L vs budget, runway, cohort retention, quality lift, hiring
Investor update	Investors	Monthly	Highlights, KPIs, burn, asks
Board pack	Board	Quarterly	Strategy, financial statements, risk register, OKR results

Dashboard specification: Appendix A18. Rule: no metric goes on a dashboard without a written definition, an owner and the decision it informs.

11

Implementation Roadmap

From the first 7 days to year 3: what to do, who owns it, what it costs, and how to know it worked.

11.1 First 7 days

11.2 Days 8-30

11.3 Days 31-60

11.4 Days 61-90

11.5 Months 4-12

11.6 Long-term initiatives

Each initiative lists its objective, actions, owner, dependencies, cost or effort, deadline, success metric, and risk with fallback. Rows marked **M** are must-have and rows marked **O** are optional. Costs are ILLUSTRATIVE.

11.1

First 7 days: essential setup

	Objective · actions	Owner	Depends on	Cost	By	Success · risk → fallback
M	**Code safety:** git init, private GitHub org, CI (typecheck, build, tests), secret scanning	CTO	–	1 day	D2	CI green on main · risk: none
M	**Legal questions out:** brief counsel on Kaggle terms, entity, trademark	CEO	–	₹15–30k	D3	Written opinion requested · slow → ask Kaggle directly
M	**Key security P0:** encrypt key columns. Design per-run tokens.	CTO	CI	3 days	D7	No plaintext keys in DB · complexity → encrypt now, tokens in sprint 2
M	**Booking page:** replace placeholder link with real scheduler	CEO	–	₹0	D2	First call booked
O	Google OAuth configured in Supabase	CTO	Google Cloud project	2 h	D5	Google sign-in works

11.2

Days 8–30: core foundations

	Objective · actions	Owner	Depends on	Cost	By	Success · risk → fallback
M	**Production launch:** deploy, domain, Supabase paid plan with backups, error tracking	CTO	Week 1	₹5k/mo	D21	Site live, uptime above 99.5% · `/api/plan` timeout → background job
M	**Incorporate + accounts:** entity, bank, accounting software, GST decision	CEO	Counsel	₹20–40k	D30	Books open · delays → operate as founder until done, no paid plans
M	**Discovery:** 10 interviews (Business ICP)	CEO	Booking page	Time	D25	Pain validated 6 of 10 · weak → change segment
M	**Quality fix:** skill replay in train-mode retrain rounds. Canary run.	CTO	CI	3 days	D20	Identity ≥ 95% across rounds
O	Onboarding checklist UI for Connect	Product	Design contractor	₹40k	D30	Connect drop-off −25%

11.3

Days 31–60: implementation and validation

	Objective · actions	Owner	Depends on	Cost	By	Success · risk → fallback
M	**Payments:** Razorpay plans, top-ups, credits, webhooks, GST invoices	CTO	Entity, GST	2 weeks	D50	Test payment end-to-end · KYC delay → manual invoices for Business
M	**3 Business pilots** delivered with acceptance tests	CEO	Discovery	Time + compute	D60	2 of 3 pass and convert · fail → learn, refine offer
M	**Launch campaign** (Chapter 7 weeks 5–8)	CEO/Growth	Production	₹45k	D56	150 sign-ups, 40 trained models
O	Hire #1 engineer (offer out)	CEO	Funding view	₹1.2L/mo	D60	Offer accepted

11.4

Days 61–90: optimisation and measurable growth

	Objective · actions	Owner	Depends on	Cost	By	Success · risk → fallback
M	**Hosting beta (≤3B):** gateway, keys, metering, multi-LoRA serving	CTO	Payments, GPU provider	3 weeks + GPU	D85	10 hosted models, p95 first token under 1.5 s · cost too high → Kaggle-only plus Business
M	**Fundraise:** deck, data room, 20 angel conversations	CEO	Pilots, metrics	Time	D90 term sheet	₹25L committed · no → month-5 rule (§3.8)
O	Paid search experiment	Growth	Activation over 30%	₹50k	D90	CAC per activated user under ₹400

11.5

Months 4–12: scaling and maturity

Initiative	Owner	Target month	Success metric
M · Dataset generation + document upload	CTO	M5	Median quality lift +10 points vs. dataset-only runs
M · Paid L40S training	CTO	M6	Margin ≥ 55% per GPU-hour
M · Success hire + Business playbook	CEO	M6	5 live clients, CSAT ≥ 4.5
M · DPDP readiness	CEO + counsel	M9	Notice, rights process, DPAs done before May 2027
O · Agency programme	Growth	M8	3 active agencies
O · Indian-language models	CTO	M10	Hindi support bot passes acceptance tests
O · Website chat widget	CTO	M9	50% of Business clients embed it

11.6

Long-term strategic initiatives (years 2–3)

- **Private deployments** for enterprises (models served inside the customer's cloud).
- **Model marketplace:** users can share or sell their tested specialist models.
- **Continuous learning:** corrections from live chats flow automatically into nightly fix runs, with human approval.
- **International expansion** to price-sensitive markets with the same "own your model" positioning.

Templates & Appendices

Twenty ready-to-use templates. Copy them into the team's docs space and fill them in.

- A1-A4 Strategy and finance
- A5-A9 Product and engineering
- A10-A14 Growth, sales and support
- A15-A20 Operations and risk
- Bibliography
- Implementation checklist

A1

Company strategy one-pager

Mission	Give every business and maker a language model of their own that does one job well.
Customer	Indian SMB support teams (first), indie developers, agencies, educators
Problem	Custom AI needs ML skills, GPUs and evaluation. Generic bots are not on-brand or on-topic.
Solution	Describe → train free on own GPU → test → publish → chat/API → fix
Why us / now	Working tested pipeline. Free compute model. Small open models are now good enough.
Business model	Paid training (GPU-hours), hosting plans, Business done-for-you
North star	Models in active use
12-month goals	₹1.9L MRR (base case) · 5 Business clients · 80 hosting customers
Top risks	Kaggle terms · model quality · cash · key security
This quarter's bets	Launch safely · 3 pilots · payments · hosting beta · raise ₹25L

A2

Annual and monthly budget

Account	M1	M2	M3	…	M12	Year	Owner
6010 Salaries	50,000	50,000	50,000	…	1,70,000		CEO
6020 Contractors	0	40,000	40,000	…	40,000		Product
6100 Software & infra	8,000	10,000	12,000	…	12,000		CTO
6200 Marketing	10,000	25,000	40,000	…	40,000		Growth
6300 Legal & professional	40,000	20,000	15,000	…	15,000		Finance
5xxx Cost of revenue	2,000	3,000	5,000	…	65,000		CTO

ILLUSTRATIVE Fill in the year column from the full 12 months. Variances over 10% and ₹20,000 are explained monthly.

A3

Chart of accounts

See §3.2 for the complete chart (codes 1000–6900). Keep it in the accounting software. Add accounts only through the finance owner, never ad hoc.

A4

Monthly financial reporting pack

- [ ] Bank, gateway and card reconciliations complete. Clearing account at zero.
- [ ] Accruals for GPU/API usage. Deferred revenue moved. Credits liability updated.
- [ ] P&L vs budget with variance notes. Balance sheet. Cash flow.
- [ ] Tie-outs: closing cash = bank · retained earnings movement = net profit · AR ageing = receivables balance
- [ ] KPIs: MRR, gross margin, burn, runway, CAC, churn
- [ ] One-page memo: what happened, why, what changes next month

A5

Product requirements document (PRD) and bug report

Title / owner / date	e.g. "Document upload → training data" · Product · 2026-11-10
Problem	Who has it, evidence (interviews, data)
Success metric	e.g. median quality lift +10 points on runs using uploads
Scope / out of scope	What ships in v1. What explicitly doesn't.
User stories	As a… I want… so that…
Acceptance criteria	Given / When / Then, including failure cases
Design	Flows, screens, empty, loading and error states
Technical notes	APIs, schema changes, security review needed?
Rollout	Flag, beta users, metrics to watch, rollback

Bug report

Summary	One line: what is wrong where
Severity	S1–S4 (§6.5)
Steps to reproduce	Numbered. Include run id / kernel / model tag.
Expected vs actual	Plus screenshots or log tail
Environment	Browser, account type, Gradio/job version

A6

Technical architecture specification

Context	Problem, constraints (T4 memory, Kaggle limits, cost)
Proposal	Components, diagram, data flow
Alternatives	At least two, with reasons rejected
Security and privacy	Data touched, keys, threats, mitigations
Operability	Monitoring, alerts, runbook, cost estimate
Migration and rollback	Steps, reversibility

A7

API documentation template

```
## POST /v1/chat  (proposed hosting API)
Auth:     Authorization: Bearer zb_…
Body:     { "model": "you/support-bot@v3",
            "messages": [{ "role": "user", "content": "…" }],
            "stream": true, "max_tokens": 1024 }
200:      text/event-stream of { "delta": "…" } … { "done": true, "usage": {…} }
Errors:   401 invalid_key · 402 quota_exceeded · 404 model_not_found
          409 model_offline · 429 rate_limited (Retry-After) · 500 internal
Limits:   Starter 10 req/min, 1 concurrent · Pro 30 req/min, 3 concurrent
```

A8

Database schema template

```
create table public.hosted_models (
  id uuid primary key default gen_random_uuid(),
  user_id uuid not null references auth.users(id) on delete cascade,
  model text not null, tag text not null, plan text not null,
  status text not null default 'provisioning',
  created_at timestamptz default now(), unique (user_id, model, tag));
alter table public.hosted_models enable row level security;
create policy own on public.hosted_models for all using (auth.uid() = user_id);
```

Every new table: primary key, owner column, RLS on, an owner-only policy, indexes for its main query, and a retention note.

A9

Design system specification

Component	Name, purpose, when not to use
Anatomy	Parts and tokens used (colour, type, spacing, radius)
Variants and states	Default, hover, focus, active, disabled, loading, error
Accessibility	Role, keyboard, labels, contrast, motion
Responsive	Behaviour at 360 / 768 / 1280
Code	Path, props, example

A10

Marketing campaign brief

Campaign	Name, dates, owner
Objective + KPI	e.g. 150 sign-ups, 40 trained models
Audience	Segment (§7.2) and where they are
Message + proof	One line and three proof points
Channels + assets	List with due dates
Budget	By line
CTA + tracking	URL with UTM, event names
Review	Date and decision rule (continue / change / stop)

A11

Content calendar

Week	Mon	Wed	Fri	Owner
1	LinkedIn: proof post (scores)	Blog: how-to guide	Short video: before/after	Growth
2	Build-in-public update	HF showcase model + card	Newsletter	CEO
3	Customer question of the week	Use-case landing page	Workshop promo	Growth
4	Monthly "models people made"	Case study	Retro: what worked	Growth

A12

Sales outreach sequence

Day 1 personalised email · day 4 video with their questions · day 9 case study · day 16 break-up email (full text in §8.4). Log every touch in the CRM, and stop on any reply or opt-out.

A13

CRM pipeline

Deal	Segment	Stage	Value / mo	Prob.	Next step	Date	Owner
Example Store ILLUSTRATIVE	D2C	4 Pilot proposed	₹25,000	50%	Review SOW call	—	CEO

A14

Customer support SOP

1. **Intake:** support email or in-app form → ticket with plan, run id, model.
2. **Categorise:** account/keys · run failure · model quality · billing · other.
3. **Respond within SLA** (Business 4 business hours, Starter/Pro 1 business day, Free best effort).
4. **Resolve:** known fixes in the help centre. Run failures follow the triage SOP (§2.4). Quality issues are routed to Fix/Testify.
5. **Escalate:** to the CTO if it's a platform bug (S1/S2 opens an incident), to the CEO if it's a commercial dispute.
6. **Close:** confirm with the customer, send CSAT, tag the root cause. Review tags weekly.

A15

Incident response checklist

- [ ] Declare: severity (§6.5), incident lead, channel, start time
- [ ] Contain: revoke or rotate affected keys, disable the feature by flag, block abusive accounts
- [ ] Assess: what data or users are affected and since when (audit logs)
- [ ] Communicate: status page, affected users. For personal-data breaches, counsel advises on DPDP notification.
- [ ] Recover: fix, verify, monitor for 24 h
- [ ] Review within 5 days: blameless post-mortem, actions with owners and dates

A16

Hiring scorecard

Competency	Evidence sought	Score 1-4	Notes
Role skill (e.g. PyTorch/PEFT, TypeScript)	Work sample: improve a failing Zebrain test or a run		
Ownership	Story of shipping something end to end		
Clear writing	Short written exercise		
Customer empathy	Explain a technical failure to a shop owner		
Values fit (§1.1)	Examples of honesty about limits, frugality		

A17

Weekly executive report

Headline	One sentence: the week in one line
North star	Models in active use: value, Δ vs last week
Funnel	Visitors · sign-ups · activated · paying
Runs	Started · published · failure causes (top 3)
Revenue	MRR · new · churned · pipeline coverage
Cash	Balance · runway months
Wins / misses	3 each
Next week's top 3	Outcome, owner

A18

KPI dashboard specification

Tile	Metric (§10.1)	Chart	Filter	Refresh
1	Models in active use	Line, 12 weeks	Plan, base model	Daily
2	Funnel conversion	Bar by step	Week, source	Daily
3	Run success by failure stage	Stacked bar	Mode	Hourly
4	MRR and net new MRR	Line + bars	Plan	Daily
5	Gross margin, burn, runway	Numbers with sparkline	Month	Monthly
6	Quality lift per version	Box plot	Use case	Daily

A19

Risk register

ID	Risk	Category	L	I	Score	Mitigation	Owner	Review date	Status
R1	Kaggle blocks free tier	Platform	3	5	15	§9.4	CEO	Monthly	Open
R2	Credential leak	Security	2	5	10	§4.10	CTO	Monthly	Open
R3…									

A20

Product launch checklist

- [ ] Legal: ToS, Privacy Policy, refund policy published
- [ ] Kaggle-use legal opinion received
- [ ] Keys encrypted, per-run tokens live
- [ ] Backups + restore drill done
- [ ] Payments tested end to end, GST invoices correct
- [ ] Monitoring, alerts, status page
- [ ] Support inbox + SLAs + help centre (top 15 questions)
- [ ] Analytics events firing. Dashboard tiles populated.
- [ ] Onboarding emails scheduled
- [ ] 3 showcase models with model cards
- [ ] Launch posts, video, Product Hunt page ready
- [ ] Booking page live (placeholder link replaced)
- [ ] Rollback plan and on-call for launch week
- [ ] Post-launch review date set (launch + 14 days)

B

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C

Final implementation checklist

> **Reminder:** every ILLUSTRATIVE number in this manual is hypothetical, and every UNKNOWN item in §0.2 must be confirmed before acting on the sections that depend on it.

Must-have before public launch

- [ ] Kaggle-use legal opinion (§9.4)
- [ ] Entity, bank, books, GST position (§3, §9.1)
- [ ] Git, CI, production deploy, backups (§4.9)
- [ ] Encrypted keys + per-run tokens (§4.10)
- [ ] ToS, Privacy Policy, refund policy (§9.5)
- [ ] Payments with GST invoices (§11.3)
- [ ] Train-mode skill replay fix (§4.5)
- [ ] Support SOP + SLAs (A14)
- [ ] Analytics + weekly report (§10, A17)

Optional / next

- [ ] Google sign-in
- [ ] Onboarding checklist UI
- [ ] Hosting beta (≤3B)
- [ ] Dataset generation + document upload
- [ ] Paid L40S training
- [ ] Agency programme
- [ ] Indian-language models
- [ ] Logo and brand guideline
- [ ] Drop or finish the unused token tables